

Nonlinear Binscatter Methods*

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Abstract

Binscatters are a powerful tool for empirical work in the social, behavioral, and biomedical sciences. Available tools rely on least squares estimation of the conditional mean. We introduce novel binscatter methods based on nonlinear, possibly nonsmooth M-estimation, covering generalized linear, robust, and quantile regression models. We provide theoretical results and practical tools, including optimal bin selection, confidence bands, and statistical tests regarding functional form or shape restrictions. We demonstrate our methods by studying the relationship of income and (lack of) health insurance. We provide software for `Python`, `R`, and `Stata`. Our technical results may be of independent interest.

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1 Introduction

Data visualization is a crucial step in any statistical analysis. In the era of big data it has become increasingly important to have simple yet informative visual tools to guide, supplement, or in some cases even supplant, numerical statistical analyses. However, it is important to maintain statistical formality and rigor to ensure the validity of any conclusions based on the data. We seek to accomplish both of these goals—effective visualization couched in a formal framework—with binned scatter plot methods.

Often known simply as a *binscatter*, the binned scatter plot has become a popular tool for visualization in large data sets, particularly in the social and behavioral sciences. The goal is to flexibly estimate, and visualize, features of the conditional distribution of a scalar outcome y_i , which may be discrete or continuous, given a covariate or treatment variable x_i , which is scalar and continuous, while possibly also controlling for a d -dimensional vector of additional factors $\mathbf{w}_i$. For estimating the conditional mean, as in the traditional regression analysis, binning has a long history: so familiar is this approach that over 60 years ago [Tukey \(1961\)](#), calling it a *regressogram*, went so far as to claim that “[a]ll statisticians who handle data know how to attack the simple case of this situation where y and x are both single real numbers” (p. 682). Going on to describe the construction, Tukey writes: “[t]he x -axis is to be divided into suitable intervals, the mean of all the y -values corresponding to x -values falling in each given interval is to be found, the results are then to be plotted, either as points, each located above the center of the corresponding x -interval, or better as horizontal bars, each extending over the corresponding x -interval” (p. 682).

This construction (perhaps disappointingly often with plotted dots rather than horizontal lines) has gained popularity in statistics, economics, and data science. The prevalence of binscatter plots can be partly ascribed to its intuitive construction and compelling visualization properties: given only data on x_i and y_i , the plot is a clean and interpretable depiction of the conditional mean. Moreover, several limitations of the classical scatter plot account for the rising use of binned scatter plots in modern analyses. First, in big data sets the classical

scatter plot is too dense to be informative, particularly about general “patterns” in the data which are to be modeled in subsequent analyses. Second, somewhat conversely, in cases where privacy is a concern the scatter plot is not allowed, regardless of its informational use as a visualization tool. Third, classical scatter plots do not provide a well-defined way to control for other factors, a common goal in treatment effect estimation and causal inference. Finally, particularly relevant to our setting, a scatter plot is not useful when outcomes are discrete. In contrast, a *binned* scatter plot provides a simple, yet flexible way of visualizing features of the conditional distribution of a (possibly discrete) outcome variable given a continuous covariate (or treatment) of interest, while controlling for other important factors.

Formally, a binscatter is grounded in the classical semilinear regression model. To date, however, binscatters have been available only to visualize (and estimate) conditional mean functions fitted using least squares. A common usage in this setting is comparing the nonparametric estimate to a linear fit, as a precursor to linear regression analysis. See [Starr and Goldfarb \(2020\)](#) for a practical review and background references. In the least squares setting, a binscatter is formally an estimator of a semilinear model for the conditional mean, nonparametric in the covariate of interest and linear in the controls, where the nonparametric component is estimated by partitioned regression. [Cattaneo et al. \(2024\)](#) used that framework to derive formal statistical properties of canonical binscatter, including correcting a common mistake in empirical practice when using controls, and provide asymptotically valid confidence bands and optimal tuning parameter selection.

The restriction to least squares semilinear regression to estimate the conditional mean has limited the applicability of binscatter methods. For one, important features of the data, such as spread or variability, cannot be visualized. Further, existing methods (and theory) can be misleading in settings where the outcome is discrete or otherwise restricted. For example, in the empirical illustration we use throughout, we study uninsuredness rates using a fractional outcome model, most naturally fitted using quasi-likelihood methods based on the logistic link. Last but not least, binscatter methods for quantile regression analysis are currently lacking in

the literature, despite their usefulness for empirical work.

This paper introduces and studies a broad class of binscatter M-estimation methods allowing for (i) a nonlinear and/or nonsmooth loss function and (ii) a nonlinear link function. Our results provide for the use of binned scatter plots for various visualization goals and different data types, particular leading cases being semiparametric conditional quantile regression and generalized partially linear models. We make several methodological and theoretical contributions: (i) we propose a feasible method for optimal tuning parameter selection to choose the appropriate number of bins; (ii) we provide (pointwise and) uniform inference procedures to construct confidence bands and hypothesis tests for parametric specifications and shape restrictions, and (iii) we develop group-wise comparisons for continuous treatment effects or for treatment effect heterogeneity. Developing these methods relies on novel technical work: allowing for a large class of binning methods, including random binning, we prove new uniform (in x) Bahadur representations and strong approximations, and thus uniform distribution theory, for the broad class of nonlinear semiparametric models considered. Obtaining these results for nonlinear, nonsmooth models, with data-dependent partitions and (a growing number of) additional controls, represents the main technical contributions of our paper, some of which may be of independent interest. For example, our results may be useful for first-step nonparametric estimation in larger inference tasks, which often involves nonlinear models.

Our nonlinear binscatter methods help restore, and in cases such as discrete outcomes or additional controls, surpass, the utility of the conventional scatter plot. We offer principled ways to visually assess patterns in the data, quantify uncertainty, and test hypotheses about the findings. Our results on quantile regression allow researchers to assess the spread of the conditional distribution, detect outliers or influential observations in the data, and study a larger class of treatment effects, formalizing and expanding common practices based on the classical scatter plot in small data, while controlling for additional factors. Our confidence bands properly quantify and communicate the uncertainty around the estimated function of interest, and can also be used to guide further analyses. We also develop formal uniform hypothesis

testing procedures regarding those functions, to assess shape constraints and parametric specifications. Causal inference is an important application area of our uniform inference results: studying treatment effect heterogeneity for binary treatments or the dose response function for a continuous treatment without imposing a functional form (e.g., to evaluate important hypotheses such as monotonicity in the dosage).

The paper proceeds as follows. We next discuss the connections between our work and the existing literature. Section 2 introduces binned scatter plots, defines the statistical model, and clarifies the parameters of interest. Section 3 gives details on our theoretical contributions and states our most important technical results. Applied-minded readers can skip this section. Then, Sections 4 and 5 build on this theory to deliver tuning parameter selection and uniform inference, respectively. Section 6 concludes.

Throughout the paper we illustrate our methods and results with a running empirical application using zip code-level data from the American Community Survey (ACS). The dependent variable, y_i , is the percentage of individuals without health insurance, and the independent variable of interest, x_i , is per capita income. These data are natural for our purposes because the presence of Medicaid, and other known economic/empirical regularities, tell us in advance how different features of the data should look, and so, rather than aim for novel empirical findings, we illustrate how our methods recover these known features.

The Supplemental Appendix (SA hereafter) contains additional technical and implementation details, all proofs, and further discussion of how our technical contributions improve on the literature. General-purpose software in `Python`, `R`, and `Stata` and replication files are available at <https://github.com/nppackages/binsreg>; see Cattaneo et al. (2025).

1.1 Related Literature

This paper contributes to several strands of the literature. First, our work builds upon and extends existing binned scatter plot methods available for applied research. Binned scatter plots are particularly popular in applied microeconomics, but have become commonly used

in a wide variety of disciplines. Examples of applications include research in climate studies (Schlenker and Roberts, 2009), health economics (Agarwal et al., 2019), marketing (Shapiro, 2020), urban economics (Su, 2022), and finance (Gathergood and Olafsson, in press). Starr and Goldfarb (2020) and Cattaneo et al. (2024) give further examples. All these existing applications are linear binscatter methods used least squares loss. This motivates our work, as our main methodological contribution is to introduce nonlinear binscatter methods, constructed using a general, possibly nonsmooth semilinear M-estimation approach. As a result, we propose a broad array of new binscatter methods for generalized linear models (e.g., Logit or Probit), robust semiparametric regression (e.g., Huber), and quantile regression.

Second, our theoretical results contribute to the literature on series/sieve estimation in general, and partitioning-based methods in particular (i.e., piecewise polynomials and splines). See Györfi et al. (2002) for a textbook introduction, and Huang (2003), Cattaneo and Farrell (2013), Belloni et al. (2015), Cattaneo et al. (2020), as well as references therein, for prior convergence rates and distribution theory. These prior works studied uniform estimation and inference for linear piecewise polynomials and spline series regression without data-driven partitioning and without additional covariates, often imposing strong regularity conditions. Our primary technical contributions as compared to the recent literature are (i) allowing for general, possibly nonlinear and nonsmooth M-estimation, (ii) allowing for random partitions and hence random basis functions in the series estimator, (iii) controlling for a diverging number of other factors in a semilinear model, and (iv) obtaining novel strong approximations and uniform inference under weaker conditions than those previously available. A substrand of the series estimation literature studies quantile regression, the closest antecedent to our work being Belloni et al. (2019). Unlike that prior work, we consider general nonlinear, possibly nonsmooth, M-estimation problems and allow for random partitions and additional controls, and our technical results are obtained under weaker regularity conditions, which in particular permit the use of piecewise constant fitting necessary for a binned scatter plot. Notably, none of the results in Cattaneo et al. (2024) are applicable to the large class of nonlinear binscat-

ter estimators considered in this paper, because they only consider least squares semilinear binscatter models. Section 3 below expands on how our theoretical techniques and results are novel relative to the extant literature. Even more detail is given throughout the SA.

Finally, our paper also contributes to the literature on data visualization, which has become an increasingly active field of study in recent years due to the rise of big data and machine learning methods. Our results speak directly to this literature, and in particular to the need for clear and explicit depictions of uncertainty, both in terms of variance and of estimation error (Healy, 2018). These are crucial in data visualization in science and research contexts as this “builds trust and credibility” (Schwabish, 2021, p.189).

2 Setup

A binned scatter plot is designed to provide a flexible, nonparametric estimate of a regression function. The construction and interpretation of a binned scatter plot is simple and intuitive, which drives their appeal for applied work. But as we will see, there are some subtleties when binned scatter plots are applied to nonlinear, nonsmooth models, especially with controls.

To describe the construction, it is helpful to first make precise the model and objects of interest. Our goal is to learn a regression function (which need not be the conditional mean) that features in the conditional distribution of a scalar outcome y_i , which may be discrete or continuous, given a covariate or treatment variable x_i , which is scalar and continuous, while possibly also controlling for a d -dimensional vector of additional factors $\mathbf{w}_i$. In applications, the goal is to flexibly study the relationship of y_i to x_i , but not necessarily to discover (or allow for) heterogeneity or nonlinearity in $\mathbf{w}_i$. The set of controls $\mathbf{w}_i$ often contains factor variables, and can be larger in dimension (d may diverge with n as made precise below). We assume the regression function depends on the scalar index $\theta_0(x_i, \mathbf{w}_i) := \mu_0(x_i) + \mathbf{w}_i' \boldsymbol{\gamma}_0$, for an unknown function μ_0 and vector $\boldsymbol{\gamma}_0$, and is thus partially linear in nature.

The model is defined by the following structure, which determines how the scalar index

$\theta_0(x_i, \mathbf{w}_i)$ relates to the outcome y_i . Let θ be a generic value of the index. For a loss function $\rho(y; \eta(\theta))$ and an inverse link function $\eta(\theta)$, let

$$(\mu_0(\cdot), \gamma_0) = \arg \min_{\mu \in \mathcal{M}, \gamma \in \mathbb{R}^d} \mathbb{E}[\rho(y_i; \eta(\mu(x_i) + \mathbf{w}_i' \gamma))], \quad (2.1)$$

where we assume the solution is unique, the model is correctly specified, and $\rho(y; \eta(\theta))$, $\eta(\theta)$, and the function class $\mathcal{M}$ obey typical boundedness and smoothness restrictions discussed in Section 3. For different choices of $\rho(\cdot)$ and $\eta(\cdot)$ this formulation covers a large class of problems including generalized linear models, robust regression, quantile regression, and nonlinear least squares. In the Appendix below we use the running empirical application to illustrate benefits of nonlinear modeling in the context of binscatter plots, which include robustness to outliers and corrupted data, valid predictions not exceeding the support of the outcome, and informative measures of spread distinct from uncertainty around the mean. Below are several leading examples that illustrate the breadth of possible applications of our framework. Textbook treatments of other nonlinear models and their use in economics are available in [Cameron and Trivedi \(2005, Part IV\)](#) and [Wooldridge \(2010, Part IV\)](#).

Example 1 (Least Squares Regression). Setting $\eta(\theta) = \theta$ and $\rho(y; \eta) = (y - \eta)^2$ recovers semiparametric least squares regression for partially linear models. $\lrcorner$

Example 2 (Logistic Regression). Assume that the binary outcome y_i , conditional on x_i and $\mathbf{w}_i$, is distributed Bernoulli with probability $\eta(\mu(x_i) + \mathbf{w}_i' \gamma)$, where $\eta(\theta) = (1 + \exp(-\theta))^{-1}$, then $\rho(y; \eta) = -y \log(\eta) - (1 - y) \log(1 - \eta)$. $\lrcorner$

Example 3 (Quantile Regression). Set $\rho(y; \eta) = [\tau - \mathbb{1}(y < \eta)](y - \eta)$ with $\eta(\theta) = \theta$ for a user-specified quantile $\tau \in (0, 1)$. $\lrcorner$

The key statistical challenge is (uniform in x) recovery of the function $\mu_0(x)$. Once accomplished, we can cover a wide variety of objects derived from (2.1). For concreteness we will focus on the following three objects, as they are of primary practical importance:

- (i) the level of the regression function, $\vartheta_0(x, \mathbf{w}) = \eta(\mu_0(x) + \mathbf{w}'\boldsymbol{\gamma}_0) = \eta(\theta_0(x, \mathbf{w}))$,
- (ii) the nonparametric component itself (or its derivative), $\mu_0^{(v)}(x) = \frac{d^v}{dx^v}\mu_0(x)$, $v \geq 0$, and
- (iii) the marginal effect $\zeta_0(x, \mathbf{w}) = \frac{\partial}{\partial x}\eta(\mu_0(x) + \mathbf{w}'\boldsymbol{\gamma}_0) = \eta^{(1)}(\theta_0(x, \mathbf{w}))\mu_0^{(1)}(x)$,

where $h^{(1)}(u) = \frac{d}{du}h(u)$ denotes the derivative of a function with respect to its scalar argument and $\mathbf{w}$ is a user-chosen evaluation point for the additional controls.

Most commonly, $\mathbf{w}$ is chosen as a summary statistic, such as the mean or the median or other quantile, or a pre-specified value such as the baseline category for discrete variables or fixed effects. Category-specific values or user-specified evaluation profiles may be useful for comparison across groups. We require that $\mathbf{w}$ be consistently estimable. The role of $\mathbf{w}$ in both plotting and inference introduces some important nuances that are discussed below.

Each of these parameters corresponds to different questions. The level, $\vartheta_0(x, \mathbf{w})$, is directly useful for visualization of the relationship of y_i to x_i and is commonly used in causal inference. If the variable x_i is a continuous treatment, our results yield a nonparametric estimate of the dose response function, while controlling for relevant factors $\mathbf{w}_i$. A plot of $\vartheta_0(x, \mathbf{w})$ shows this function for the subgroup defined by $\mathbf{w}_i = \mathbf{w}$. We can also obtain separate dose response functions for different subgroups of the data to be used in multi-sample comparisons. On the other hand, if x_i is a pre-treatment variable, the same multi-sample results provide an analysis of treatment effect heterogeneity for discrete (often binary) treatments, and our uniform inference allows for discovery of treatment effect heterogeneity. Finally, $\vartheta_0(x, \mathbf{w})$ in the quantile regression case can be used to assess the spread of the conditional distribution (such as the interquartile range or quantiles close to zero and one) or robust measures of central tendency, as would be done with a classical scatter plot.

The nonparametric component is most often studied to assess functional form, generally against a parsimonious parametric specification to be considered for later analyses or for a shape restriction that is of substantive interest, such as monotonicity or convexity. Historically, a common use of binscatter was to visually (and informally) assess if $\mu_0(x)$ is well-approximated

by a linear model, and if so, proceed under that specification. Our results provide rigor to such practice, and expand the idea to a much richer class of models and hypotheses.

The marginal (or partial) effect $\zeta_0(x, \mathbf{w})$ is a standard object in economic analysis of nonlinear models. In binary choice models it is common to study how the probability of $y = 1$ changes as a function of x . The marginal effect at the average, obtained by setting $\mathbf{w}$ to the sample mean, is a standard way to summarize nonlinear models by giving the effect for the “average” individual. For example, we show that the marginal effect of income changes sign in our application, indicating a changing response in the uninsuredness rate as a result of Medicaid. Comparing marginal effects across groups for heterogeneity analysis, in causal or noncausal settings, is a common goal in social science applications with nonlinear models. Note that we obtain the marginal effect at the average, but not the average marginal effect. This distinction is not present for linear binscatters (Cattaneo et al., 2024), but for nonlinear models, these differ in both implementation and interpretation (see Cameron and Trivedi (2005, §5.2.4)). It is conceivable that our results can be used as a first step, to obtain $\zeta_0(x, \mathbf{w} = \mathbf{w}_i)$, prior to integrating over the distribution of $\mathbf{w}_i$ for the average marginal effect.

2.1 Estimation

Given an i.i.d. sample $(y_i, x_i, \mathbf{w}_i')'$, $i = 1, \dots, n$, the binscatter estimator is constructed by solving the empirical analogue of (2.1) using a partitioning-based approximation to the unknown function $\mu_0(x)$. This nonparametric approximation requires two choices: the partitioning of the support of x_i and the estimation within each bin.

To fix ideas, it is useful to begin with the simplest case where local constant fitting is used and $\mathbf{w}_i$ is absent. First, the support of x_i is divided into $J < n$ disjoint bins. J is the main tuning parameter for this nonparametric estimation problem, and its choice is crucial both visually and statistically. To describe the estimator, we will take $J < n$ as given at present, and return to its choice in Section 4 below.

Coupled with a choice of J is a method to divide the support. A theoretical innovation

of our work is that the bin breakpoints themselves can be data-dependent, provided the y_i 's are not used. This is distinct from a data-driven choice of J . The partition is denoted by $\hat{\Delta} = \{\hat{\mathcal{B}}_1, \hat{\mathcal{B}}_2, \dots, \hat{\mathcal{B}}_J\}$, with the bins and breakpoints denoted by

$$\hat{\mathcal{B}}_j = \begin{cases} [\hat{\tau}_{j-1}, \hat{\tau}_j) & \text{if } j = 1, 2, \dots, J-1, \\ [\hat{\tau}_{J-1}, \hat{\tau}_J] & \text{if } j = J, \end{cases}$$

and the union of the bins equal to the support of x_i . The breakpoints $\{\hat{\tau}_j\}_{j=1}^J$ result from a user-chosen, possibly data-driven, partitioning method. Our theoretical results cover any random partition that is independent of the outcomes y_i 's (given x_i 's and $\mathbf{w}_i$'s), and “quasi-uniform”, which intuitively requires the bins to be sufficiently similar. The formal condition is stated in the next section. The simplest approach is evenly-spaced breakpoint locations ($\hat{\tau}_j = x_{\min} + j(x_{\max} - x_{\min})/J$). The most popular choice, however, is to use the empirical quantiles of x_i , setting $\hat{\tau}_j = \hat{F}^{-1}(j/J)$ with $\hat{F}(u) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}(x_i \leq u)$ the standard empirical cumulative distribution function, and $\hat{F}^{-1}$ its generalized inverse. In the SA we show that both of these satisfy our generic assumptions. Other possible methods include certain adaptive regression trees and related partitioning methods, such as those with the so-called “X-property” or via sample splitting; see [Devroye et al. \(2013\)](#), [Zhang and Singer \(2010\)](#), and references therein. For concreteness, we will use quantile spacing for empirical analysis throughout the paper.

Given a partition $\hat{\Delta}$, the binscatter estimate is formed by fitting the sample analogue of (2.1) within each bin, using only an intercept. In the simple case of least squares regression, this is identical to computing the sample average of y_i for observations in each bin, exactly as [Tukey \(1961\)](#) described, yielding a piecewise constant approximation to the unknown conditional expectation. The same method is followed for all other models. For example, in the case of binary data or fractional outcomes, a logistic regression of y_i on a constant is fit for each bin. For the median, or any other quantile, one simply computes the empirical quantile of y_i using observations only within the bin.

Formally, we define the basis functions $\widehat{\mathbf{b}}_0(x) = [\mathbb{1}_{\widehat{\mathcal{B}}_1}(x), \mathbb{1}_{\widehat{\mathcal{B}}_2}(x), \dots, \mathbb{1}_{\widehat{\mathcal{B}}_J}(x)]'$, consisting of indicators for each bin. We then obtain

$$\widehat{\mu}(x) = \widehat{\mathbf{b}}_0(x)' \widehat{\boldsymbol{\beta}}, \quad \widehat{\boldsymbol{\beta}} = \arg \min_{\boldsymbol{\beta} \in \mathbb{R}^J} \sum_{i=1}^n \rho\left(y_i; \eta(\widehat{\mathbf{b}}_0(x_i)' \boldsymbol{\beta})\right). \quad (2.2)$$

To help fix ideas, Figure 1 illustrates the basics of how a binscatter is constructed. The data are obtained from the ACS using the 5-year survey estimates beginning in 2013 and ending in 2017 (available from the Census Bureau website). All analyses are performed at the zip code tabulation area level for the United States (excluding Puerto Rico). The dependent variable, y_i , is the percentage of individuals without health insurance, and the independent variable of interest, x_i , is per capita income. The fractional nature of the outcome motivates the use of logistic quasi-maximum likelihood for estimation and inference (Papke and Wooldridge, 1996).¹ Figure 1(a) shows the classical scatter plot of the raw data. This data set has about 32,000 observations, far from the millions commonly encountered, and already this plot fails to be useful for assessing the functional form: the visualization is dominated by a dense cloud of data with a few outlying observations. Figure 1(b) shows a binned scatter plot being constructed, with the raw data in the background. The dots are the fitted values of applying (2.2) following Example 2, i.e., we show $\widehat{\vartheta}(x) = \eta(\widehat{\mu}(x))$. Figure 1(c) isolates the binscatter and overlays a linear fit (i.e., a global logistic quasi-likelihood with $\mu_0(x)$ assumed linear in x). The linear approximation to $\mu_0(x)$ appears satisfactory at first, but this is only because the nonparametric estimate is oversmoothed. We know this to be the case both *statistically*, as 10 bins is far fewer than the IMSE-optimal value (81 in this case; see Section 4), and *economically*, because the existence of Medicaid means fewer uninsured people at the lowest incomes. Figure 1(d) presents the estimate using the optimal number of bins and recovers the

¹For piecewise constant modeling of the conditional mean without controls, the fitted values from QMLE approaches will be identical to those from least squares, as both return the sample means in each bin. Since valid inference (§5 ahead) requires bias correction, the equivalence is for fitted values only. Further, this equivalence does not apply to marginal effects, quantiles, or other estimands, nor to when other basis functions are used, nor to when controls are added. The Appendix provides further discussion and examples of the benefits of nonlinear modeling, including showing that in these data, linear binscatter plots may lie outside $y_i \in [0, 1]$.

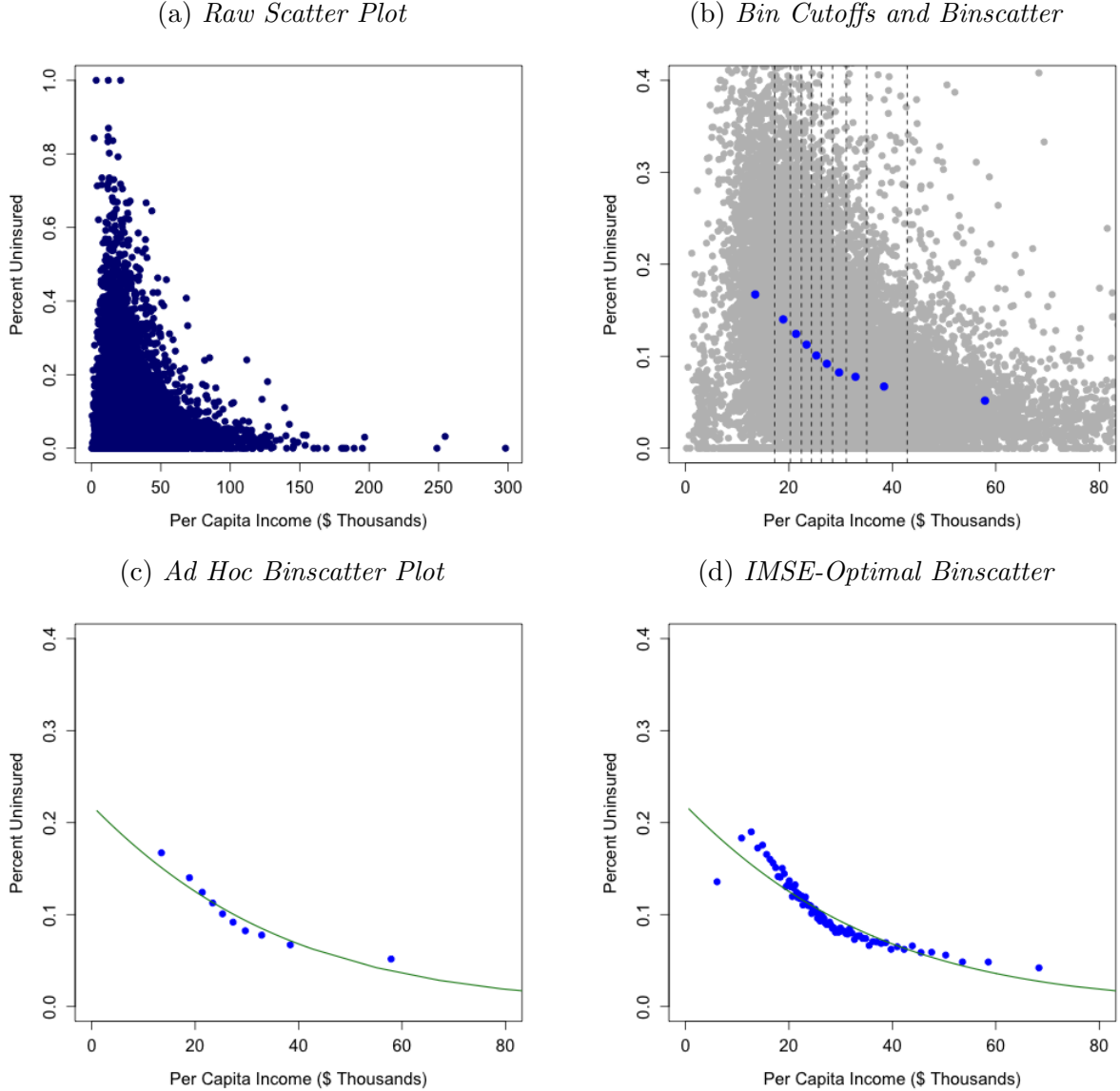
correct shape, which is missed in the informal analysis using an ad hoc choice of J . Figure 1(d) also shows the inadequacy of parametric modeling in this case, and the need for the flexibility provided by nonlinear binscatter. As a preview, Table 1 below shows that formal tests reject polynomial parametric specifications and reject the hypothesis that the uninsurance rate is monotonically decreasing with per capita income.

Graphs like Figures 1(c) and (d) have a long tradition in statistics and data science, and have recently become ubiquitous in applied microeconomics. Visually assessing functional forms is the typical use. Importantly, in this case the visualization shows an estimate of $\vartheta_0(x, \mathbf{w})$, not $\mu_0(x)$ directly. Further, although the binned scatter plot invites the viewer to “connect the dots” smoothly, the actual estimator is piecewise constant, which generally gives a less appealing visualization but underpins any formal analysis.

We expand on (2.2) in two ways: adding the covariates $\mathbf{w}_i$ and enriching the set of allowable basis functions. The covariates can be directly incorporated into the loss, exactly as they are in (2.1). Moreover, the additively separable and linear nature of the controls makes this generalization straightforward empirically. Importantly, the presence of controls invalidates bin-by-bin estimation, as the coefficients γ_0 are global parameters. The SA discusses different approaches to estimating μ_0 and γ_0 .

Next, we replace the piecewise constant approximation based on $\widehat{\mathbf{b}}_0(x)$ with an order- p polynomial approximation in each bin that is $(s - 1)$ -times continuously differentiable at the breakpoints of the bins, with the convention that $s = 0$ corresponds to discontinuous fits and $s = 1$ is continuous but nondifferentiable. This change to the basis gives additional flexibility that is crucial for bias reduction and derivative estimation, the latter being instrumental for studying shape restrictions and specification testing. By construction, $p \geq s \geq 0$, and derivative estimation requires that the derivative of interest, $v \geq 0$, is no larger than p . The general basis is defined as $\widehat{\mathbf{b}}_{p,s}(x) := \widehat{\mathbf{T}}_s[\widehat{\mathbf{b}}_0(x) \otimes (1, x, \dots, x^p)']$, where $\otimes$ denotes the Kronecker product and $\widehat{\mathbf{T}}_s$ is a $[(p + 1)J - (J - 1)s] \times (p + 1)J$ transformation matrix ensuring that the $(s - 1)$ -th derivative of the estimate is continuous ($\widehat{\mathbf{T}}_s$ depends on the random partition and

Figure 1: **Illustration of Nonlinear Binned Scatter Plots.** This figure illustrates the construction of a nonlinear binned scatter plot using the ACS data. All analyses are performed at the zip code tabulation area level for the United States (excluding Puerto Rico). The dependent variable is the percentage of individuals without health insurance and the independent variable of interest is per capita income.



its exact form is given in Section SA-5.2 of [Cattaneo et al. \(2024\)](#).

The class of bases we consider includes both piecewise polynomials as well as standard splines. We draw the distinction only to highlight that we do not necessarily enforce smoothness at the breakpoints. Textbook treatments often separate the two with the same distinction ([Györfi et al., 2002](#)). But all the bases are locally-supported and partition-based, and thus

similar, compared to, for example, global polynomials, which are ruled out. When $s = 0$, $\widehat{\mathbf{T}}_s$ is the identity matrix, and the fit is a piecewise polynomial of degree p (i.e. a separate fit within each bin). The standard binscatter uses piecewise constant fit, shown as lines or as dots, corresponding to $s = p = 0$. Another popular choice are cubic B-splines, obtained by setting $s = p = 3$. On account of its popularity and to simplify notation, we will assume throughout the paper that $s = p$ and use the notation $\widehat{\mathbf{b}}_p(x) := \widehat{\mathbf{b}}_{p,p}(x)$. The SA treats the general case of $s \leq p$. Finally, for technical reasons, $\widehat{\mathbf{b}}_p(x)$ and associated coefficients are understood to be represented in the standardized coordinates defined in SA-2. This is an invertible reparameterization of the basis defined above and leaves fitted functions and variance formulas unchanged, and thus we omit the details here.

The generalized, covariate-adjusted binscatter can now be defined. We first solve²

$$\begin{bmatrix} \widehat{\boldsymbol{\beta}} \\ \widehat{\boldsymbol{\gamma}} \end{bmatrix} = \arg \min_{\boldsymbol{\beta}, \boldsymbol{\gamma}} \sum_{i=1}^n \rho\left(y_i; \eta(\widehat{\mathbf{b}}_p(x_i)' \boldsymbol{\beta} + \mathbf{w}_i' \boldsymbol{\gamma})\right). \quad (2.3)$$

Next, using (2.3) the estimators of the three functions of interest are:

$$\widehat{\vartheta}_p(x, \widehat{\mathbf{w}}) = \eta(\widehat{\theta}_p(x, \widehat{\mathbf{w}})), \quad (2.4)$$

$$\widehat{\mu}_p^{(v)}(x) = \widehat{\mathbf{b}}_p^{(v)}(x)' \widehat{\boldsymbol{\beta}}, \quad (2.5)$$

$$\widehat{\zeta}_p(x, \widehat{\mathbf{w}}) = \eta^{(1)}(\widehat{\theta}_p(x, \widehat{\mathbf{w}})) \widehat{\mu}_p^{(1)}(x), \quad (2.6)$$

respectively, where $\widehat{\theta}_p(x, \widehat{\mathbf{w}}) = \widehat{\mu}_p(x) + \widehat{\mathbf{w}}' \widehat{\boldsymbol{\gamma}}$ is the plug-in estimator of the true index $\theta_0(x, \mathbf{w})$, and $\widehat{\mathbf{w}}$ (non-random or generated based on $\{\mathbf{w}_i\}_{i=1}^n$) is a consistent estimator for the desired evaluation point $\mathbf{w}$. We will often make the polynomial order p explicit, as this is needed for clarity when constructing confidence bands and hypothesis tests in Section 5; dependence on

²It is possible to add nonstochastic, per-observation weights to the loss function (2.3), as might be natural in aggregate data such as our running example. Under appropriate conditions on these weights, our theory will continue to hold. We do not make this explicit to save notation, but our software implementation allows certain types of weights, depending on the platform's native implementation. See <https://github.com/nppackages/binsreg> for platform-specific documentation.

$\widehat{\Delta}$ and choice of J is suppressed, but also important.

Nonlinear binscatter methods can be constructed for both fixed $J < \infty$ and large $J \rightarrow \infty$ as $n \rightarrow \infty$, naturally leading to different interpretations. The (true, nonparametric) functions of interest defined at the beginning of this section cannot be recovered when J is fixed, but coarsened versions thereof will be, and these objects can have an interesting interpretation: if the partition “settles” as $n \rightarrow \infty$ to some fixed Δ_0 with associated fixed basis $\mathbf{b}_p(x)$ (see Assumption 5 below for a precise definition), then the probability limit of the fixed- J binscatter is the solution to (2.1) where the function class $\mathcal{M}$ is restricted to be $\mathcal{M} = \{\mu(x) = \mathbf{b}_p(x)' \boldsymbol{\beta} : \boldsymbol{\beta} \in \mathbb{R}^K, K = \dim(\mathbf{b}_p(x))\}$. This is most natural with $p = 0$ and quantile-spacing, because the binscatter shows average outcomes across quantiles of a continuous covariate. For $p = 0$ and $J = 100$, for example, the results allow for comparison of y_i across percentiles of x_i (possibly controlling for $\mathbf{w}_i$), which is standard for x_i variables such as test scores or measures of wealth. All our estimation and inference results remain valid when J is fixed, provided the target parameter is adjusted accordingly and d is treated as fixed (keeping focus on the relationship of y_i and x_i being the more difficult statistical problem). See Cattaneo et al. (2024, 2025) for further discussion of fixed- J binscatter methods. Here we will consider only the case $J \rightarrow \infty$ as $n \rightarrow \infty$ to streamline the presentation and theory.

3 Theory

This section presents two main novel technical results: a uniform Bahadur representation (Theorem 1) and a feasible strong approximation (Theorem 2). The methodological results in subsequent sections—tuning parameter selection, confidence bands, and hypothesis testing—are built from these results. To conserve space and notation, in this section we only show the results for $\widehat{\vartheta}_p(x, \widehat{\mathbf{w}})$. The analogous results for $\widehat{\mu}_p^{(v)}(x)$ and $\widehat{\zeta}_p(x, \widehat{\mathbf{w}})$ are deferred to the SA (specific references below) and are conceptually similar. The SA also gives other important technical results and additional discussion of how our theory improves on the existing litera-

ture. Note that because the results in this section are technical in nature and serve as building blocks for the methodology, this section may be skipped by the applied-minded reader and we do not revisit the running empirical application here.

First we state the assumptions required on the data generating process and statistical model. These are mostly standard and not restrictive, but involve some notation and technicalities, particularly to allow for $d = \dim(\mathbf{w}_i) \rightarrow \infty$. To reduce clutter the dependence on n is left implicit (from e.g. J , d , $\mathbf{w}$, and so forth). Unless explicitly stated otherwise, all constants and all upper, lower, moment, and Lipschitz bounds in the assumptions below are uniform in n . The class of data generating processes is restricted by the following.

Assumption 1 (Data Generating Process).

- (i) $\{(y_i, x_i, \mathbf{w}'_i) : 1 \leq i \leq n\}$ are i.i.d. random vectors supported on $\mathcal{Y} \times \mathcal{X} \times \mathcal{W}$ and satisfy the correctly specified model (2.1), where $(\mu_0(\cdot), \gamma_0)$ is the unique solution, and $\sup_n \|\gamma_0\| < \infty$, where $\|\cdot\|$ is the Euclidean norm. $\mathcal{X}$ is a compact interval, $\mathcal{W}$ is a compact set, and $\sup_n \sup_{\mathbf{w} \in \mathcal{W}} \|\mathbf{w}\|_\infty < \infty$, where $\|\cdot\|_\infty$ is the sup norm.
- (ii) The marginal distribution of x_i has a Lipschitz continuous (Lebesgue) density bounded away from zero on $\mathcal{X}$.
- (iii) If the conditional distribution of y_i given $(x_i, \mathbf{w}'_i)$ has a (conditional) density with respect to Lebesgue measure, this density is uniformly bounded over its support and $\mathcal{X} \times \mathcal{W}$.

This assumption restricts attention to cross-sectional data and bounded covariates with minimal and standard regularity imposed on the underlying joint distribution. Requiring x_i to be continuously distributed is natural given the visualization and estimation goals, but our results can also be applied to discrete x_i by taking each mass point as its own bin to conduct simultaneous estimation and inference over those support points.

The next assumption gives regularity conditions on the class of statistical models.

Assumption 2 (Statistical Model).

- (i) For some $\delta > 0$, there exists a compact interval $\mathcal{I}$ contained in the interior of the domain of $\eta(\cdot)$ such that $[\theta_0(x, \mathbf{w}) - \delta, \theta_0(x, \mathbf{w}) + \delta] \subseteq \mathcal{I}$ uniformly over n , $x \in \mathcal{X}$, and $\mathbf{w} \in \mathcal{W}$, where neither δ nor $\mathcal{I}$ depend on n . $\rho(y; \eta)$ is absolutely continuous with respect to η on an open set containing $\eta(\mathcal{I})$ and admits a derivative $\psi(y, \eta) := \psi^\dagger(y - \eta)\psi^\dagger(\eta)$ almost everywhere on this set. $\psi^\dagger(\cdot)$ is continuously differentiable and strictly positive or negative on this set. If the conditional distribution of y_i given $(x_i, \mathbf{w}'_i)$ does not have a Lebesgue density, then $\psi^\dagger(\cdot)$ is Lipschitz continuous, otherwise it is piecewise Lipschitz with finitely many discontinuity points.
- (ii) $\rho(y; \eta(\theta))$ is convex with respect to θ and $\eta(\cdot)$ is strictly monotonic and thrice continuously differentiable.
- (iii) $\Psi(x, \mathbf{w}; \eta) := \mathbb{E}[\psi(y_i, \eta) | x_i = x, \mathbf{w}_i = \mathbf{w}]$ is twice continuously differentiable in η on $\eta(\mathcal{I})$ with derivatives bounded uniformly in n and $\mathcal{X} \times \mathcal{W} \times \eta(\mathcal{I})$. $\Psi(x_i, \mathbf{w}_i; \eta(\theta_0(x_i, \mathbf{w}_i))) = 0$. For $\Upsilon(x, \mathbf{w}) := \frac{\partial}{\partial \eta} \Psi(x, \mathbf{w}; \eta)|_{\eta=\eta(\theta_0(x, \mathbf{w}))}$, $\Upsilon(x, \mathbf{w})\eta^{(1)}(\theta_0(x, \mathbf{w}))^2$ is bounded away from zero uniformly over $\mathcal{X} \times \mathcal{W}$ and $\mathbb{E}[\Upsilon(x_i, \mathbf{w}_i)\eta^{(1)}(\theta_0(x_i, \mathbf{w}_i))^2 | x_i = x]$ is Lipschitz continuous on $\mathcal{X}$. $\mathbb{E}[|\psi(y_i, \eta(\theta_0(x_i, \mathbf{w}_i)))|^\nu | x_i = x, \mathbf{w}_i = \mathbf{w}]$ is uniformly bounded over $\mathcal{X} \times \mathcal{W}$ for some $\nu > 2$.
- (iv) $\sigma^2(x, \mathbf{w}) := \mathbb{E}[\psi(y_i, \eta(\theta_0(x_i, \mathbf{w}_i)))^2 | x_i = x, \mathbf{w}_i = \mathbf{w}]$ is bounded away from zero uniformly in $\mathcal{X} \times \mathcal{W}$. $\mathbb{E}[\eta^{(1)}(\theta_0(x_i, \mathbf{w}_i))^2 \sigma^2(x_i, \mathbf{w}_i) | x_i = x]$ is Lipschitz continuous.
- (v) $\mu_0(\cdot)$ is ς -times continuously differentiable for some $\varsigma \geq p + 1$ with derivatives bounded uniformly in n and $\mathcal{X}$.

This assumption imposes regularity conditions on the statistical model in (2.1), particularly on the loss function and resulting parameters of interest. The complexity of part (i) reflects the breadth of the class of models and parameters we cover. When y_i is continuous the loss function can have points of nondifferentiability, but for discrete outcomes the loss must be smoother. To illustrate, consider first Example 3 in Section 2: the loss function for quantile

regression is continuous but not differentiable everywhere, which is covered by our assumptions with $\psi(y, \eta) = \mathbb{1}(y - \eta < 0) - \tau$, where $\psi^\dagger(y - \eta) = \mathbb{1}(y - \eta < 0) - \tau$ exhibits a discontinuity point, and $\psi^\ddagger(\eta) = 1$ is smooth. Alternatively, for logistic regression (Example 2), $y_i \in \{0, 1\}$ and we have $\psi(y, \eta) = -(y - \eta)[\eta(1 - \eta)]^{-1}$, which exactly matches the required structure of $\psi^\dagger(y - \eta)\psi^\ddagger(\eta)$. Both functions are clearly as smooth as required and the definition of $\eta(\theta)$ ensures that $\psi^\ddagger > 0$. The rest of the assumption gives standard moment and boundedness conditions to ensure that the parameters and their estimators are well-defined; those regularity conditions are also satisfied in all examples of interest.

Finally, the nonparametric object $\mu_0(\cdot)$ is assumed to be smooth, as is standard in the nonparametric inference literature. Our theory assumes sufficient smoothness based on a fixed p , stemming from the viewpoint that practitioners select a polynomial degree for their estimator and then conduct estimation and inference based on that choice. This is particularly true for binscatter applications, where piecewise constant estimators are de facto standard, but also gives rise to the ubiquity of (e.g.) cubic splines for regression or local linear methods in regression discontinuity designs. Taking p as given, we assume sufficient regularity for feasible tuning parameter selection and inference ($\varsigma \geq p + 1$). In contrast, if one views ς as known and fixed, p can be chosen to be equal or larger than the smoothness assumed on $\mu_0(\cdot)$, which generally yields the fastest convergence rate by speeding up the bias rate to the smoothness-imposed limit, while only inflating the variance constant. The implausibility of knowing the exact smoothness aside, this comes at the expense of feasible tuning parameter selection. [Calonico et al. \(2022\)](#) give more discussion on assumed versus exploited smoothness and how this relates to inference in practice. In typical applied research, assumed smoothness is not binding and does not dictate the choice of estimator.

3.1 Bahadur Representation

Here we state our novel Bahadur representation. This result is essentially a stochastic linearization of the estimator uniformly in $x \in \mathcal{X}$. In addition to the assumptions stated above,

we require the following high-level conditions on the estimation procedure. These conditions ensure that the partitioning scheme is sufficiently regular and that both the coefficients and the evaluation point for the control variables can be estimated sufficiently well.

Assumption 3 (High-Level Estimation Conditions).

(i) *The partition $\hat{\Delta}$ is independent of $\{y_i\}_{i=1}^n$ given $\{x_i, \mathbf{w}_i\}_{i=1}^n$ and, with probability approaching 1, $\max_{1 \leq j \leq J} |\hat{\tau}_j - \hat{\tau}_{j-1}| \leq C \min_{1 \leq j \leq J} |\hat{\tau}_j - \hat{\tau}_{j-1}|$, for an absolute constant $C > 0$.*

(ii) *$\|\hat{\gamma} - \gamma_0\| = O_{\mathbb{P}}(r_{\gamma})$ for a deterministic sequence r_{γ} satisfying $\sqrt{d} r_{\gamma} = o(\sqrt{J/n} + J^{-p-1})$. Moreover, $\mathbf{w} = \mathbf{w}_n \in \mathcal{W}$, $\|\hat{\mathbf{w}} - \mathbf{w}\| = o_{\mathbb{P}}(1)$ and $\mathbb{P}[\hat{\mathbf{w}} \in \mathcal{W}] \rightarrow 1$.*

The requirement that the partition intervals are not too dissimilar in length is satisfied for evenly spaced partitioning, trivially, and is shown to hold for quantile spacing in the SA (Lemma SA-5.2). For other data-driven methods this condition must be checked. This assumed property of the random binning structure is often called quasi-uniformity (Huang, 2003; Cattaneo et al., 2020), and is important for controlling the approximation bias and, when combined with the assumptions on the density of x_i , for ensuring that each bin contains sufficient data to control the variance. Much more discussion is available in the SA.

Part (ii) requires that the coefficient vector γ_0 can be estimated sufficiently accurately and the desired evaluation point of $\mathbf{w}_i$ can be estimated consistently. The latter is not restrictive in general. Any summary statistic, such as the mean, minimum, maximum, or a specific quantile, is feasible for $\mathbf{w}$. Prespecified values, such as categories for discrete variables, require no estimation and trivially satisfy this condition. Combining the two, which might serve for category-specific binscatters, is also allowed. For estimation of γ_0 a rate is required. For fixed (or very slowly-diverging) d this is not generally restrictive, as the nonparametric estimation of $\mu_0(x)$ is the most statistically difficult part of the problem. However, when $d \rightarrow \infty$, estimation becomes more difficult for the coefficients and the stated convergence requirement interacts with choices of J and p and ν in Assumption 2. We do not cover the case where

$\mathbf{w}_i$ contains individual-specific intercepts as fixed effects, such as in panel data, which would require d proportional to n : in all cases $d = o(n)$. The most popular implementation of local constant estimation ($p = 0$) with an IMSE-optimal choice of $J = J_{\text{IMSE}} \asymp n^{1/3}$ (see Theorem 3 below) is allowed if $\nu > 6$ and $d = o(n^{1/6}/\sqrt{\log n})$. For the remainder of the main text we maintain the high-level condition above for notational simplicity, as this covers all cases of fixed and diverging dimension. Section SA-3.9 gives complete details, including verification of the assumption under primitive conditions and discussion of different rate restrictions.

We can now give our first theoretical result: a uniform (in x) Bahadur representation for $\widehat{\vartheta}_p(x, \widehat{\mathbf{w}})$, as defined in (2.4).

Theorem 1 (Bahadur Representation). *Let $\ell_n := \log n + \sqrt{d \log n}$. Suppose that Assumptions 1, 2, and 3 hold, and*

$$\frac{d + \log n}{J} + \frac{J^{\frac{\nu}{\nu-2}} \log n}{n} + \frac{J^{\frac{2\nu}{\nu-1}} (\log n)^{\frac{\nu}{\nu-1}}}{n} = o(1), \quad \frac{J \ell_n^2}{n^{1-1/\nu}} \lesssim \sqrt{\frac{J \log n}{n}} + J^{-p-1}.$$

Then,

$$\sup_{x \in \mathcal{X}} |\widehat{\vartheta}_p(x, \widehat{\mathbf{w}}) - \vartheta_0(x, \mathbf{w}) - \widehat{\mathbf{L}}_p(x, \mathbf{w})| \lesssim_{\mathbb{P}} r_n + \|\widehat{\mathbf{w}} - \mathbf{w}\|,$$

where the infeasible Gram matrix $\bar{\mathbf{Q}}_p := n^{-1} \sum_{i=1}^n \widehat{\mathbf{b}}_p(x_i) \widehat{\mathbf{b}}_p(x_i)' \Upsilon(x_i, \mathbf{w}_i) \eta^{(1)}(\theta_0(x_i, \mathbf{w}_i))^2$,

$$\widehat{\mathbf{L}}_p(x, \mathbf{w}) := -\eta^{(1)}(\theta_0(x, \mathbf{w})) \widehat{\mathbf{b}}_p(x)' \bar{\mathbf{Q}}_p^{-1} \frac{1}{n} \sum_{i=1}^n \widehat{\mathbf{b}}_p(x_i) \eta^{(1)}(\theta_0(x_i, \mathbf{w}_i)) \psi(y_i, \eta(\theta_0(x_i, \mathbf{w}_i))),$$

and

$$r_n := \left(\frac{J \log n}{n} \right)^{3/4} \ell_n + J^{-\frac{p+1}{2}} \sqrt{\frac{J}{n}} \ell_n + \frac{J \ell_n^2}{n} + J^{-p-1} + \sqrt{d} r_\gamma.$$

This result is one fundamental building block for the strong approximation in the next section. Further, this directly yields important consequences including the mean squared error expansion used to choose J and the asymptotic variance formula for inference. This will be key in our results and is discussed below. The analogous Bahadur representations for $\widehat{\mu}_p^{(v)}(x)$ and $\widehat{\zeta}_p(x, \widehat{\mathbf{w}})$, under the same assumptions, are given in Theorem SA-3.1. The “linear”

term is slightly different to account for the different structure of the three estimands and the remainder rate for derivative estimation is slower.

Theorem 1 improves on the econometric literature in several ways. Here we give a short summary, but further information is available in the SA (Remark SA-3.3), wherein we also discuss the contributions of several preliminary lemmas, which may be of separate interest in the series estimation literature. First, this result substantially generalizes the least squares results in Cattaneo et al. (2024), under essentially the same rate restrictions, with an error of approximation that is optimal up to $\log(n)$ terms.

Second, compared to other work in the series estimation literature, our results allow for random partitioning schemes and do not require that the partition be “convergent” in any sense, which allows for general partitioning schemes. It is not possible to obtain the uniform results of Theorem 1 (or the strong approximations below) by expanding around a nonrandom limit. Thus $\widehat{\mathbf{b}}_p^{(v)}(x)$ is left as random in the Bahadur representation, including in the matrix $\bar{\mathbf{Q}}_p$, which are infeasible but not population objects.

Third, we require only weak rate restrictions. Specifically, when d is fixed, we require $J^{\frac{8}{3}}/n = o(1)$ up to $\log n$ terms when $\nu \geq 4$ (see Assumption 2), thus accommodating IMSE-optimal (Theorem 3 ahead) piecewise constant binscatter estimators which are ruled out by prior work. In fact, for piecewise polynomials ($s = 0$), we can show that the Bahadur representation still holds under $J/n = o(1)$ up to $\log n$ terms when a subexponential moment restriction holds for the “score” $\psi(y_i, \eta_i)$, which is analogous to the results of Kong et al. (2010) for kernel-based estimators. Belloni et al. (2019) considers series-based quantile regression, and Theorem 2 and Corollary 2 therein can be used to establish a Bahadur representation and uniform convergence of the resulting estimators under (the substantially stronger) $J^4/n^{1-\varepsilon} = o(1)$ for some $\varepsilon > 0$. However, Kong et al. (2010) handle weakly dependent data, which we do not, and Belloni et al. (2019) obtain a result uniform in both $x \in \mathcal{X}$ and in the quantile index (over a compact set), whereas our result is for a specific quantile. Our theory could be extended in both directions.

We use several novel theoretical techniques to achieve these improvements. First, unlike

Cattaneo et al. (2024) where the least-squares estimators have explicit closed-form expressions, our proof derives the linear representation directly from the M-estimation loss and its scores, allowing for nonlinear links and possibly nonsmooth loss functions. Second, relative to prior general sieve theory, we exploit the locally supported, sparse binscatter basis and the exponential off-diagonal decay of the inverse Gram matrix to localize each coefficient perturbation used in the objective comparison to a small neighborhood; this accommodates random partitions and avoids the additional full-dimensional complexity penalty from treating all J basis coefficients as generic regressors, thereby yielding weaker rate restrictions.

3.2 Strong Approximation

With the Bahadur representation in place, we can develop tools for inference. Our main result is a strong approximation for the (Studentized) t -statistic process for each of the three estimators, allowing us to obtain a feasible asymptotic distributional approximation. Again we give the details only for $\widehat{\vartheta}_p(x, \widehat{\mathbf{w}})$ and defer the others to the SA.

The variance is an immediate consequence of the expansion in Theorem 1, and is made feasible by replacing unknown objects by their estimators. The form of the variance is reminiscent of its parametric counterpart (e.g., for generalized linear models), but estimation is more complicated. Herein we give a general high-level condition that the infeasible Gram matrix can be estimated sufficiently well, complementing the conditions of Assumption 3, which can justify several alternatives commonly used in practice. These are discussed in Section SA-4.1. See Section SA-4.1 for examples of $\widehat{\Upsilon}(x_i, \mathbf{w}_i)$ for different models.

Assumption 4 (High-Level Variance Estimation Condition). *For the infeasible Gram matrix $\bar{\mathbf{Q}}_p$ defined in Theorem 1, there is an estimator $\widehat{\Upsilon}(x_i, \mathbf{w}_i)$ such that $\|\bar{\mathbf{Q}}_p - \widehat{\mathbf{Q}}_p\|_o = O_{\mathbb{P}}(J^{-p-1} + (\frac{J \log n}{n^{1-2/\nu}})^{1/2})$, where $\widehat{\mathbf{Q}}_p := n^{-1} \sum_{i=1}^n \widehat{\mathbf{b}}_p(x_i) \widehat{\mathbf{b}}_p(x_i)' \widehat{\Upsilon}(x_i, \mathbf{w}_i) \eta^{(1)}(\widehat{\theta}_p(x_i, \mathbf{w}_i))^2$ and $\|\cdot\|_o$ is the operator norm.*

With the Gram matrix estimator in place, we define the t -statistic as follows, for a fixed p

as a process over $x \in \mathcal{X}$.

$$T_{\vartheta,p}(x) = \frac{\widehat{\vartheta}_p(x, \widehat{\mathbf{w}}) - \vartheta_0(x, \mathbf{w})}{\sqrt{\widehat{\Omega}_{\vartheta,p}(x)/n}}, \quad \widehat{\Omega}_{\vartheta,p}(x) := \eta^{(1)}(\widehat{\theta}_p(x, \widehat{\mathbf{w}}))^2 \widehat{\mathbf{b}}_p(x)' \widehat{\mathbf{Q}}_p^{-1} \widehat{\Sigma}_p \widehat{\mathbf{Q}}_p^{-1} \widehat{\mathbf{b}}_p(x), \quad (3.1)$$

where $\widehat{\Sigma}_p := n^{-1} \sum_{i=1}^n \widehat{\mathbf{b}}_p(x_i) \widehat{\mathbf{b}}_p(x_i)' \psi(y_i, \eta(\widehat{\theta}_p(x_i, \mathbf{w}_i)))^2 \eta^{(1)}(\widehat{\theta}_p(x_i, \mathbf{w}_i))^2$ and $\widehat{\mathbf{Q}}_p$ is defined in Assumption 4. The t -statistics $T_{\mu^{(v)},p}(x)$ and $T_{\zeta,p}(x)$, for $\mu_0^{(v)}(x)$ and $\zeta_0(x, \mathbf{w})$ respectively, are entirely analogous, and are defined in Section SA-3.3.

Our inference results will follow from the next key theorem.

Theorem 2 (Strong Approximation). *Suppose that the conditions of Theorem 1 and Assumption 4 hold, and let $(a_n : n \geq 1)$ be a positive deterministic sequence bounded away from zero such that*

$$\frac{J(\log n)^2}{n^{1-\frac{2}{\nu}}} + \left(\frac{J(\log n)^3 \ell_n^4}{n} \right)^{1/2} + nJ^{-2p-3} + \frac{\ell_n^2}{J^{p+1}} + \frac{ndr_\gamma^2}{J} = o(a_n^{-2}),$$

and $\|\widehat{\mathbf{w}} - \mathbf{w}\| = o_{\mathbb{P}}(a_n^{-1} \sqrt{J/n})$. Then, on a properly enriched probability space, there exists a $(J+p)$ -dimensional standard Normal random vector $\mathbf{N}$ such that for any $\xi > 0$,

$$\mathbb{P}\left(\sup_{x \in \mathcal{X}} |T_{\vartheta,p}(x) - \bar{Z}_{\vartheta,p}(x)| > \xi a_n^{-1}\right) = o(1), \quad \bar{Z}_{\vartheta,p}(x) = \frac{\widehat{\mathbf{b}}_p(x)' \eta^{(1)}(\theta_0(x, \mathbf{w})) \bar{\mathbf{Q}}_p^{-1} \bar{\Sigma}_p^{1/2}}{\sqrt{\bar{\Omega}_{\vartheta,p}(x)}} \mathbf{N},$$

where $\bar{\Sigma}_p$ and $\bar{\Omega}_{\vartheta,p}(x)$ are shown in Section SA-3. On a further enriched space, there exists a conformable standard Normal vector $\mathbf{N}^*$, independent of $\{(y_i, x_i, \mathbf{w}_i')'\}_{i=1}^n$ and $\widehat{\Delta}$. In the following display, $\bar{Z}_{\vartheta,p}(\cdot)$ denotes a conditionally equal-in-distribution copy driven by the same $\mathbf{N}^*$. Then, for any $\xi > 0$,

$$\mathbb{P}\left(\sup_{x \in \mathcal{X}} |\bar{Z}_{\vartheta,p}(x) - \widehat{Z}_{\vartheta,p}(x)| > \xi a_n^{-1} \mid \{(y_i, x_i, \mathbf{w}_i')'\}_{i=1}^n, \widehat{\Delta}\right) = o_{\mathbb{P}}(1),$$

$$\widehat{Z}_{\vartheta,p}(x) = \frac{\widehat{\mathbf{b}}_p(x)' \eta^{(1)}(\widehat{\theta}_p(x, \widehat{\mathbf{w}})) \widehat{\mathbf{Q}}_p^{-1} \widehat{\Sigma}_p^{1/2}}{\sqrt{\widehat{\Omega}_{\vartheta,p}(x)}} \mathbf{N}^*.$$

The approximating process, $\bar{Z}_{\vartheta,p}(\cdot)$, is a Gaussian process conditional on $\{x_i, \mathbf{w}_i\}_{i=1}^n$ and $\hat{\Delta}$ by construction, and the elements of $\bar{\Sigma}_p$, $\bar{\Omega}_{\vartheta,p}(x)$ and $\hat{\mathbf{b}}_p(x)$ reflect this conditioning. This process is infeasible but the second result shows that all the unknown quantities in $\bar{Z}_{\vartheta,p}(\cdot)$ can be replaced by their sample analogues to obtain a feasible approximation. Theorems SA-3.5 and SA-3.6 give the corresponding results for $T_{\mu^{(v)},p}(x)$ and $T_{\zeta,p}(x)$, under the same assumptions. Pointwise inference results are also given in the SA for completeness.

Compared to the existing literature on uniform nonparametric, Theorem 2 gives several important improvements. It inherits all the novelty discussed at the close of Section 3.1 above, which is not duplicated here. The closest antecedents are Belloni et al. (2019) and Cattaneo et al. (2024). Again, Theorem 2 substantially generalizes the least squares results in Cattaneo et al. (2024), under essentially the same rate restrictions, with an error of approximation that is optimal up to $\log(n)$ terms.

Belloni et al. (2019) considers strong approximations of more general series estimators, but restricts attention to quantile regression (though again, uniform in the quantiles). We focus on partitioning-based methods, but consider a broader class of models. We have weaker assumptions than would be found by applying their results to binscatter estimators. Their Theorem 11 can be applied to construct strong approximation of the t -statistic process based on pivotal coupling that achieves the approximation rate $a_n = n^{-\varepsilon'}$ under $J^4/n^{1-\varepsilon} = o(1)$ for some constants $\varepsilon, \varepsilon' > 0$, whereas their Theorem 12 can be used to construct strong approximation based on Gaussian processes under $J^5/n^{1-\varepsilon} = o(1)$. In contrast, when d is fixed, for $a_n = \sqrt{\log n}$ and $\nu \geq 4$, we require $J^{\frac{8}{3}}/n = o(1)$, up to $\log n$ terms. Prior work has obtained such sharp results, under weak conditions, only for the supremum of the process (Chernozhukov et al., 2014), whereas our theory yields an approximation for the entire t -statistic process. The coupling technique employed here is the same as in Cattaneo et al. (2024), once the leading linear term is obtained from our Bahadur representation. Specifically, conditional on the covariates and the random partition, and after ordering observations by x_i , we couple scalar partial sums with Gaussian partial sums and exploit the local binscatter

structure, rather than directly approximating an increasing-dimensional coefficient vector as in generic series methods. This approach accommodates random partitions and delivers weaker rate restrictions than prior general series methods.

4 Tuning Parameter Selection

We can use our theoretical results from the previous section to directly inform implementation. Our first task is selecting the number of bins, J , which determines both the visual and statistical properties of the estimator. Consistent estimation and valid inference is possible for a range of diverging sequences of J , but this does not provide sufficiently precise guidance for implementation. Thus, our first methodological result is a Nagar-type integrated mean squared error expansion that enables an optimal choice of J in empirical applications.

We need one further assumption to characterize the leading terms of the expansion. Intuitively, we require that the random partition $\hat{\Delta}$ “converges” to a fixed one which obeys the same restrictions as in Assumption 3. This assumption is not needed for any other results.

Assumption 5. *There exists a non-random partition $\Delta_0 = \{\mathcal{B}_1, \dots, \mathcal{B}_J\}$ with $\mathcal{B}_j = [\tau_{j-1}, \tau_j)$ for $j \leq J-1$ and $\mathcal{B}_J = [\tau_{J-1}, \tau_J]$ such that $\max_{1 \leq j \leq J} |\tau_j - \tau_{j-1}| \leq C \min_{1 \leq j \leq J} |\tau_j - \tau_{j-1}|$ for an absolute constant $C > 0$ and $\max_{0 \leq j \leq J} |\hat{\tau}_j - \tau_j| = o_{\mathbb{P}}(J^{-1})$. For $p > 0$, also assume $\max_{2 \leq j \leq J} |(\tau_j - \tau_{j-1}) - (\tau_{j-1} - \tau_{j-2})| = o(J^{-1})$.*

This condition trivially holds for well-behaved nonrandom partitions, but also holds for the leading case of quantile-spacing, since sample quantiles converge to population counterparts. In more general cases this condition could fail. However, all our other results remain valid, and furthermore, even if this condition fails a rule-of-thumb choice of J is available, which has the optimal rate but suboptimal constants. See Section SA-4.2.

Our IMSE result for $\hat{\vartheta}_p(x, \hat{\mathbf{w}})$ is given by the following. The corresponding results for $\hat{\mu}_p^{(v)}(x)$ and $\hat{\zeta}_p(x, \hat{\mathbf{w}})$ are stated in Theorem SA-3.4.

Theorem 3. Set $\omega(x)$ to be a continuous weighting function over $\mathcal{X}$ bounded away from zero. Suppose that the conditions of Theorem 1 and Assumption 5 hold, and $\varsigma \geq (p+2)$. Assume $\|\widehat{\mathbf{w}} - \mathbf{w}\| = o_{\mathbb{P}}(\sqrt{J/n} + J^{-p-1})$, $J(\log n)^3 \ell_n^4 = o(n)$, and $\ell_n^2 = o(J)$. Then

$$\int_{\mathcal{X}} \left(\widehat{\vartheta}_p(x, \widehat{\mathbf{w}}) - \vartheta_0(x, \mathbf{w}) \right)^2 \omega(x) dx = \mathbf{AISE}_{\vartheta} + o_{\mathbb{P}}\left(\frac{J}{n} + J^{-2(p+1)}\right)$$

where, for nonrandom terms $\mathcal{V}_n(p)$ and $\mathcal{B}_n(p)$ shown in Theorem SA-3.4 that are bounded and nonzero in general, $\mathbf{AISE}_{\vartheta}$ obeys

$$\mathbb{E}[\mathbf{AISE}_{\vartheta} | \{(x_i, \mathbf{w}'_i)\}_{i=1}^n, \widehat{\Delta}] = \frac{J}{n} \mathcal{V}_n(p) + J^{-2(p+1)} \mathcal{B}_n(p) + o_{\mathbb{P}}\left(\frac{J}{n} + J^{-2(p+1)}\right)$$

This result is stated in terms of J , as it is the tuning parameter, but the rates and constants depend on the fixed polynomial order p (recall that $p = s$, see Section 2.1). An L_2 convergence rate immediately follows from this result. An L_{∞} convergence rate is available in Corollary SA-3.1. This theorem, and the L_2 and L_{∞} rates, are new to the literature, even in the case of non-random partitioning and without covariate adjustment, for nonlinear series estimators and binscatter methods. The SA contains detailed discussions of our contributions.

The practical consequence of Theorem 3 is that we can balance the (squared) bias and variance to obtain an IMSE-optimal choice of J , which is given by

$$J_{\text{IMSE}}(p) := \left(\frac{2(p+1)\mathcal{B}_n(p)}{\mathcal{V}_n(p)} \right)^{\frac{1}{2p+3}} n^{\frac{1}{2p+3}}. \quad (4.1)$$

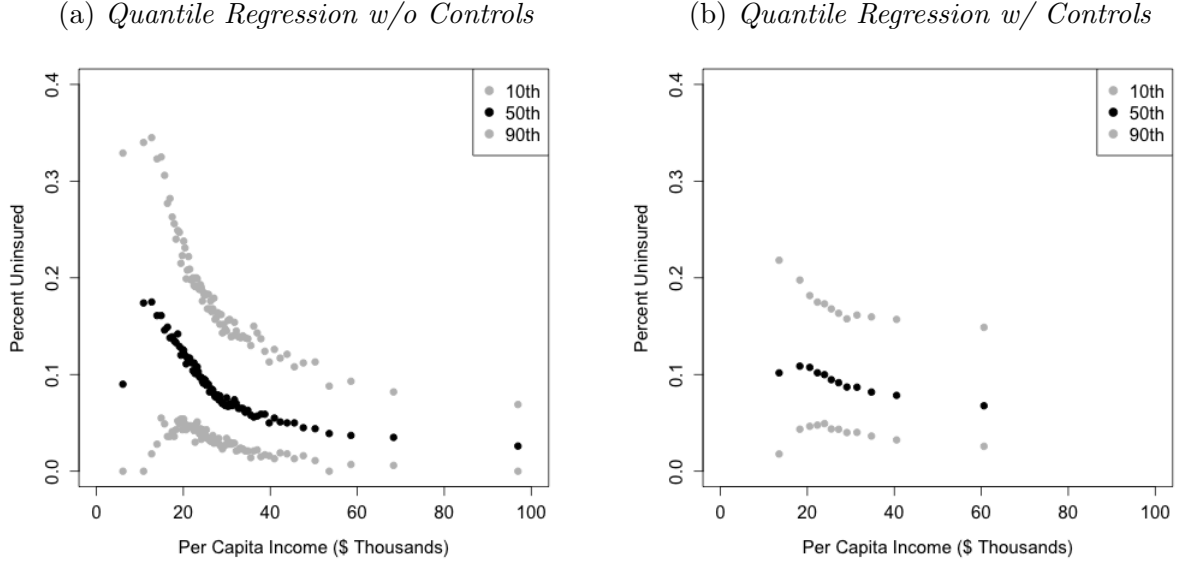
Implementing binscatter with this J is optimal in the sense of providing the IMSE-optimal estimate of the unknown function $\vartheta_0(\cdot, \mathbf{w})$. In the next section we discuss the use of $J_{\text{IMSE}}(p)$ for inference, where, as is typical, a bias correction must be applied. A feasible version of $J_{\text{IMSE}}(p)$ is described in Section SA-4 and implemented in the `binsreg` package (Cattaneo et al., 2025). Section SA-4.3 discusses binned scatter plots with a fixed choice of J , which can be visually appealing and interpretable in some applications.

Figure 2 demonstrates the use of the optimal J for quantile estimation (see Figure 1(d) for the mean). Figure 2(a) shows the quantiles of uninsuredness conditional on per capita income, while Figure 2(b) adds demographic controls following the linear structure of Equation (2.1).³ Quantile regression can be used to visualize the spread of the conditional distribution. Observe that Figure 2 restores the visualization of the dispersion in the data that is present in Figure 1(a) but hidden by the averaging in Figure 1(c). This (conditional) dispersion is distinct from uncertainty around the mean estimate, shown in Figure 3 ahead and further in the Appendix. Figure 2(a) shows that there is much larger spread in the fraction uninsured in lower income areas, but Figure 2(b) shows that this gap narrows when controlling for demographics ($\hat{\mathbf{w}}$ is set to the sample mean).⁴ Although our empirical exercise primarily serves to illustrate our new methodology and visualization, uncovering this spread at low incomes may have policy implications that warrant further study. Although Medicaid is designed to reach low-income individuals, the spread identified by the quantile curves in Figure 2 shows that there is substantial inequality that may be important for policy makers. For example, this could arise from barriers to access that differ across regions (states administer separate Medicaid programs) or across demographics, or inequality in the provision of healthcare facilities. These inequalities could further be crucial in designing targeted subsidies to better approach universal coverage (Finkelstein et al., 2019). See Koenker (2017) for a comprehensive discussion of quantile models and their empirical relevance.

³Beginning with Hall and Mishkin (1982), it has become commonplace in the literature on dynamics of income and consumption to (linearly) residualize the outcome of interest with respect to a set of demographic controls prior to analyzing nonlinear dynamics using quantile methods, to focus on the unpredictable aspects of the outcome. For a recent example see Arellano et al. (2017). The analogue in our setting would be to apply the simple binscatter of Equation (2.2), with quantile loss, using residualized $\hat{y}_i$ and $\hat{x}_i$. While such an approach may be sensible in time series applications, in our cross-sectional context this practice is unlikely to be appropriate and is difficult to interpret in general. See Cattaneo et al. (2024) for discussion and examples. The partially linear structure of $\theta_0(x_i, \mathbf{w}_i) = \mu_0(x_i) + \mathbf{w}_i' \boldsymbol{\gamma}_0$ naturally has the interpretation of “controlling for $\mathbf{w}_i$ ” which obviates any need for preprocessing of the data in our setting.

⁴We add nine controls: percentage of residents with a high school degree, percentage of residents with a bachelor’s degree, median resident age, the local unemployment rate, mean household size, percentage of residents without internet access, percentage of residents in the armed forces, percentage of residents speaking only English, and percentage of residents aged 65 and older.

Figure 2: **Conditional Quantiles.** This figure illustrates the use of quantile regression to visualize the spread in the ACS data (see Example 3). As the link function is the identity, panel (a) shows estimates of $\vartheta_0(x, \mathbf{w})$ for $\tau = 0.1, 0.5, 0.9$, while panel (b) shows the same quantiles including additional covariates controlling for demographics listed in footnote 4.



5 Uniform Inference

We now turn to uniform inference for the three functions defined in Section 2. Uniform inference is required to make statistical statements about the functions $\vartheta_0(x, \mathbf{w})$, $\mu_0^{(v)}(x)$, and $\zeta_0(x, \mathbf{w})$, rather than about their values at a specific point x . Pointwise inference methods (e.g. confidence intervals) will not suffice for our main applications of interest, including treatment effect heterogeneity and continuous treatment effects, as well as shape restrictions and functional forms. For completeness, pointwise inference results are given in the SA and implemented in the `binsreg` package, but omitted here to save space.

A key element of our uniform inference results—from a practical point of view—is the pairing of a feasible tuning parameter choice with valid inference. To give the most accurate estimate, and therefore also visualization, of $\vartheta_0(x, \mathbf{w})$ we prefer to use $J_{\text{IMSE}}(p)$ of (4.1). However, as is typical for nonparametric problems, the (I)MSE-optimal tuning parameter choice delivers invalid inference, as it fails to eliminate a first-order bias. We therefore use robust bias correction to ensure that $J_{\text{IMSE}}(p)$ remains a valid choice across all uses, delivering optimal

estimation and valid inference. With this eye toward practicality, we state the result below at the IMSE-optimal rate $J \asymp n^{1/(2p+3)}$, which includes $J_{\text{IMSE}}(p)$ in the nondegenerate case under Assumption 5, while the SA gives the more general result under mild rate restrictions on J .

Bias correction involves estimating, and removing, the leading smoothing bias term, and is made “robust” by correcting the standard errors to account for the additional sampling variability that has been introduced. Robust bias correction has theoretically superior higher-order inference properties (Calonico et al., 2018, 2022), performs well in simulations, and has been empirically validated in specific contexts (Hyytinen et al., 2018; De Magalhães et al., 2025). Robust bias correction is operationalized in the present context by (i) selecting a degree p and creating a partition $\hat{\Delta}$ based on $J_{\text{IMSE}}(p)$ to form the optimal point estimate of $\hat{\vartheta}_p(x, \hat{\mathbf{w}})$ and then (ii) conducting inference using $T_{\vartheta, p+1}(x)$ (or its feasible analogue $\hat{Z}_{\vartheta, p+1}(x)$), i.e. the statistic formed using a higher-degree polynomial but the partitioning scheme based on p in (i): $\hat{\Delta} = \hat{\Delta}(J_{\text{IMSE}}(p))$. Any higher-degree polynomial may be used, but $p + 1$ is simple and robust. Cattaneo et al. (2020) give further discussion of robust bias correction in the context of partition regression, including alternative strategies.

5.1 Confidence Bands

Our first uniform inference result delivers confidence bands for the functions $\vartheta_0(x, \mathbf{w})$, $\mu_0^{(v)}(x)$, and $\zeta_0(x, \mathbf{w})$. Confidence bands are similar in spirit to the more familiar concept of confidence intervals, but instead cover the entire function (uniformly over $x \in \mathcal{X}$) with a prespecified probability. Confidence bands are the appropriate tool for visualizing the uncertainty around the estimated function. The size of the band also changes to reflect the presence of heteroskedasticity in the data. These bands can be used directly to identify interesting or important features of the function, for example, regions where it is statistically indistinguishable from zero or from a constant function. Bands are also useful for assessing the functional form or shape, such as regions of linearity or monotonicity, and therefore visually complement the formal hypothesis tests we introduce below.

The confidence bands are built from Theorem 2, coupled with robust bias correction, as discussed above. Confidence bands are defined as the area between an upper and lower bounding function. Recall that we employ robust bias correction, so that $\widehat{\vartheta}_{p+1}(x, \widehat{\mathbf{w}})$ is the bias-corrected version of $\widehat{\vartheta}_p(x, \widehat{\mathbf{w}})$, and is thus the “center” of the confidence band, and using $\widehat{\Omega}_{\vartheta, p+1}(x)$ in the standard error accounts for the additional variability. The robust bias-corrected confidence band for $\vartheta_0(x, \mathbf{w})$ is given by

$$\widehat{I}_{\vartheta, p+1}(x) = \left[\widehat{\vartheta}_{p+1}(x, \widehat{\mathbf{w}}) \pm \mathbf{c}_{\vartheta} \sqrt{\widehat{\Omega}_{\vartheta, p+1}(x)/n} \right], \quad (5.1)$$

where the critical value is determined by

$$\mathbf{c}_{\vartheta} = \inf \left\{ c \in \mathbb{R}_+ : \mathbb{P} \left[\sup_{x \in \mathcal{X}} |\widehat{Z}_{\vartheta, p+1}(x)| \leq c \mid \{(y_i, x_i, \mathbf{w}'_i)\}_{i=1}^n, \widehat{\Delta} \right] \geq 1 - \alpha \right\}. \quad (5.2)$$

The validity of this confidence band follows from Theorem 2, and inherits the novelty discussed above, which allows us to approximate the distribution of the supremum of $T_{\vartheta, p+1}(x)$ by applying the same functional to $\widehat{Z}_{\vartheta, p+1}(x)$. This yields the following result. Here we assume directly that J has the IMSE-optimal rate. Theorem SA-3.8 states a more general result, valid for a range of J , as well as inference results for $\mu_0^{(v)}(x)$ and $\zeta_0(x, \mathbf{w})$.

Theorem 4. *Set $J \asymp n^{1/(2p+3)}$. Suppose that the conditions of Theorem 2 hold, with $p+1$ in place of p and $a_n = \sqrt{\log J}$. Then, for $\widehat{I}_{\vartheta, p+1}(x)$ defined in (5.1),*

$$\mathbb{P} \left[\vartheta_0(x, \mathbf{w}) \in \widehat{I}_{\vartheta, p+1}(x), \text{ for all } x \in \mathcal{X} \right] = 1 - \alpha + o(1).$$

This result establishes valid confidence bands for generalized, covariate-adjusted binscatters. We can use this result to visually assess uncertainty about the form and shape of the regression function. One can visually “test” hypotheses of interest, though formal testing (Section 5.2) is recommended. Plotting both the estimate $\widehat{\vartheta}_p(x, \widehat{\mathbf{w}})$ and the band $\widehat{I}_{\vartheta, p+1}(x)$ is advisable in applications because doing so presents both the IMSE-optimal point estimate and a valid

measure of uncertainty (and one that uses the same bins).

In nonlinear models, particularly in social sciences, partial effects are often the preferred way of summarizing the relationship (causal or not) of x_i to y_i , controlling for $\mathbf{w}_i$. Here, this corresponds to the estimate of $\zeta_0(x, \mathbf{w})$ and its associated confidence band. When x_i is a treatment variable, $\zeta_0(x, \mathbf{w})$ captures the effect of increasing the treatment dosage, and the band can help identify regions of $\mathcal{X}$ with the largest effects, or any other noteworthy shape.

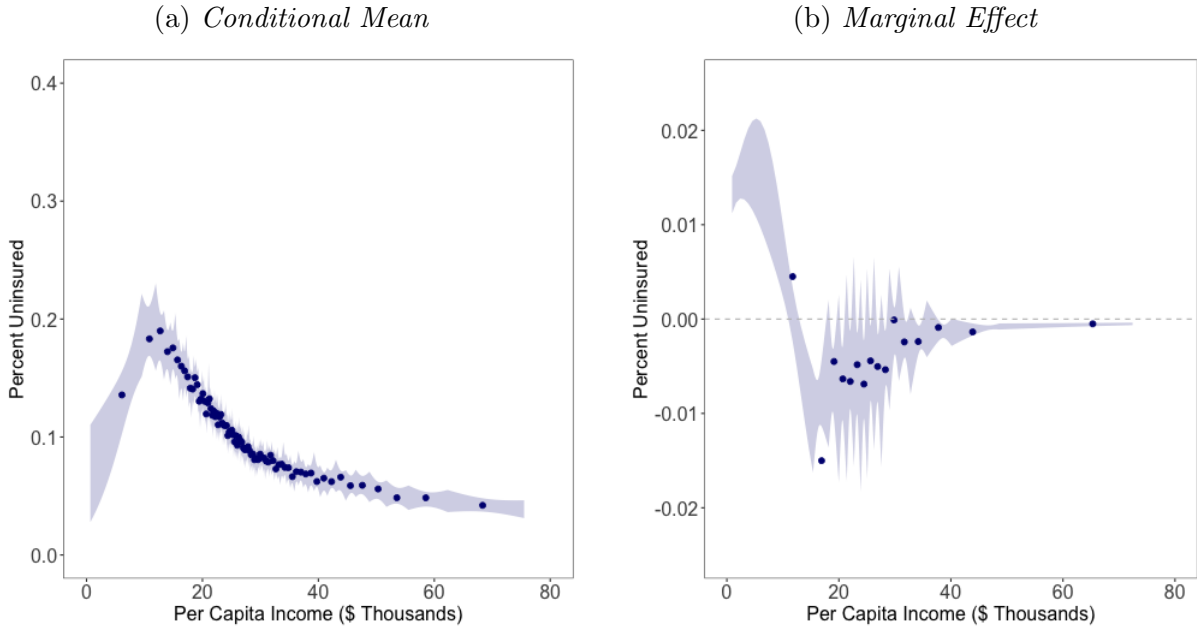
Figure 3 shows examples of confidence bands using our running empirical application. The confidence band in Figure 3(a) displays the uncertainty surrounding the estimate first shown in Figure 1(d). The presence of the Medicaid program is clearly delineated by the shape of the band at lower income levels. From the band we can immediately conclude that the relationship is nonmonotonic. This is further emphasized in Figure 3(b), showing the marginal effect. Using the bands, we can reject the null hypothesis of monotonicity as the band lies completely on either side of zero at low and high income levels.

There are two features of our confidence bands that warrant mention. First, the user-selected point of evaluation $\mathbf{w}$ can impact the shape, placement, and size, of the band. One might expect that since the additional controls are modeled as additively linear, the evaluation point $\mathbf{w}$ (and the coefficient γ_0) should not impact conclusions about the nonparametric relationship between y and x . But this intuition overlooks the fact that the function $\mu_0(x)$ is only defined relative to how $\mathbf{w}_i$ is coded. For example, if $\mathbf{w}_i$ contains a binary variable indicating groups of different sizes, then the uncertainty will be different between the two groups. For $\vartheta_0(x, \mathbf{w})$ and $\zeta_0(x, \mathbf{w})$, the shape of the confidence bands may change with the choice of $\mathbf{w}$. This impacts all aspects of inference, both visual and the formal tests below. This is not particular to our method; it is always present in analyses of models like (2.1).

Second, the bias correction may result in the point estimate lying outside the confidence band. This is formally correct but can be visually unappealing. Figure 3(b) shows an example of this phenomenon. This occurs in areas of high bias, and may arise in any application of bias correction. Being based on a polynomial of order $p + 1$, the confidence band removes

more bias than does the point estimate. This is not necessarily a failing: the point estimate remains IMSE-optimal and inference remains valid, and, importantly, both are based on theory-founded partitioning choices. Adding bias correction to the point estimate itself would render it no longer optimal, and further, would require additional ad-hoc tuning parameter choices, potentially spoiling the formal guarantees of the current approach.

Figure 3: **Confidence Bands.** This figure illustrates confidence bands in a nonlinear binned scatter plot using the ACS data. Panel (a) shows the point estimate (dots) and robust bias corrected confidence band (shaded region) for the conditional mean function with no controls, i.e., $\vartheta_0(x) = \eta(\mu_0(x))$, while panel (b) shows the corresponding point estimate and confidence band for the marginal effect, $\zeta_0(x)$. Shaded regions denote 95% confidence bands and are based on 50,000 random draws.



5.2 Hypothesis Testing

We also provide formal hypothesis tests for substantive questions including functional form or shape restrictions for $\vartheta_0(x, \mathbf{w})$, $\mu_0(x)$, and $\zeta_0(x, \mathbf{w})$. Our discussion here is brief. Full details are given in the SA.

A leading case is testing a parametric form for $\mu_0(x)$, where under the null there exists some finite-dimensional parameter $\boldsymbol{\theta}$ such that $\mu_0(x) = m(x; \boldsymbol{\theta})$, uniformly in $x \in \mathcal{X}$ (we can also

test any derivative of $\mu_0(x)$). The two-sided testing problem is

$$\begin{aligned}\dot{H}_0^\mu : \quad & \sup_{x \in \mathcal{X}} |\mu_0(x) - m(x; \boldsymbol{\theta})| = 0, \quad \text{for some } \boldsymbol{\theta}, \quad \text{vs.} \\ \dot{H}_A^\mu : \quad & \sup_{x \in \mathcal{X}} |\mu_0(x) - m(x; \boldsymbol{\theta})| > 0, \quad \text{for all } \boldsymbol{\theta}.\end{aligned}$$

This test formalizes the notion of visual inspection in plots like Figure 1(c) and (d), beyond what is already done by adding a confidence band. We test this hypothesis using the statistic

$$\dot{T}_{\mu, p+1}(x) := \frac{\hat{\mu}_{p+1}(x) - m(x; \tilde{\boldsymbol{\theta}})}{\sqrt{\hat{\Omega}_{\mu, p+1}(x)/n}},$$

where $\tilde{\boldsymbol{\theta}}$ and $\tilde{\boldsymbol{\gamma}}$ are estimators of $\boldsymbol{\theta}$ and $\boldsymbol{\gamma}_0$ under $\dot{H}_0^\mu$, where $\theta_0(x, \mathbf{w}) = m(x; \boldsymbol{\theta}) + \mathbf{w}'\boldsymbol{\gamma}_0$. Conditions on the estimators are mild, and indeed, satisfied when obtaining $\tilde{\boldsymbol{\theta}}$ and $\tilde{\boldsymbol{\gamma}}$ by fitting the hypothesized parametric model. For example, in testing a parametric model for $\mu_0(x)$, $\tilde{\boldsymbol{\theta}}$ must obey $\sup_{x \in \mathcal{X}} |\mu_0(x) - m(x; \tilde{\boldsymbol{\theta}})| = o_{\mathbb{P}}(\sqrt{J/(n \log n)})$. Section SA-3.7 gives complete details. Theorem 2 again provides the tools to obtain the appropriate critical value. Theorem SA-3.9 gives the formal result showing size control and consistency of the test, as well as the corresponding tests for $\vartheta_0(x, \mathbf{w})$ and $\zeta_0(x, \mathbf{w})$, which require further, but similarly mild, regularity conditions. Last, we also provide for testing shape restrictions, see Theorem SA-3.10, which are conceptually similar but are generally one-sided testing problems. A leading example would be testing monotonicity of $\vartheta_0(x, \mathbf{w})$ or $\zeta_0(x, \mathbf{w})$.

Table 1 shows several testing examples using the L_∞ norm. Consider first the left column of results. We test against the linear specification, formalizing the visual comparison in Figure 1(d). Here $\tilde{\boldsymbol{\theta}}$ and $\tilde{\boldsymbol{\gamma}}$ are obtained by fitting this linear index specification parametrically. We also test against a cubic (in x) logistic quasi-likelihood model for added flexibility, again obtaining $\tilde{\boldsymbol{\theta}}$ and $\tilde{\boldsymbol{\gamma}}$ from this parametric model. Both parametric specifications are rejected, and moreover, also rejected when including other controls (listed in footnote 4). This highlights the need for nonparametric modeling in this application. Finally we test the substantive null

hypothesis that the uninsuredness rate is monotonically decreasing with income. This null is also strongly rejected due to the existence of Medicaid. This motivates the right column of results, where we repeat the analysis after restricting the sample to zip codes with per capita income above 138% of the 2013–2017 average federal poverty line for a single-person household (\$16,248). This is the cutoff for expanded Medicaid eligibility based only on income. When we restrict to this sample, which diminishes the influence of the Medicaid program, we fail to reject the null hypothesis of a monotonic decline, but still reject the parametric specifications. This aligns with the need for flexible estimation and matches the conclusions we draw from the shape of the confidence bands shown in Figure 3.

Table 1: **Specification and Shape Testing**

	Full Sample			Above Income Cutoff		
	Test Stat.	p -value	$\hat{J}_{\text{IMSE}}$	Test Stat.	p -value	$\hat{J}_{\text{IMSE}}$
Test of Linear Fit						
No Covariates	22.319	0.000	81	21.344	0.000	40
Covariates, $\hat{\mathbf{w}} = \bar{\mathbf{w}}$	7.176	0.000	12	4.265	0.000	8
Test of Cubic Fit						
No Covariates	9.961	0.000	81	75.426	0.000	40
Covariates, $\hat{\mathbf{w}} = \bar{\mathbf{w}}$	9.014	0.000	12	3.506	0.001	8
Test of Monotonic Decline						
No Covariates	7.155	0.000	16	1.095	0.944	10
Covariates, $\hat{\mathbf{w}} = \bar{\mathbf{w}}$	5.929	0.000	11	0.803	0.973	9

Notes. This table reports the test statistics and associated p -values along with the IMSE-optimal choice of J from hypothesis tests of parametric specifications and shape restrictions using the ACS data. The first and second panels report test results for the null hypotheses of linear and cubic (in x) logistic quasi-likelihood models, respectively, while the third panel reports results for the null hypothesis of a monotonic decline in the level (i.e., negative derivative). All tests are performed with and without control variables (controls are listed in footnote 4). The left panel (“Full Sample”) reports results for the full sample whereas the right panel (“Above Income Cutoff”) restricts to the sample of zip codes with per capita income above \$16,248. All p -values are based on 50,000 random draws.

5.3 Multi-Sample Comparisons

Our results extend to comparisons between different samples, or groups, within the data. This is a common goal in program evaluation and causal inference settings. With discrete

(e.g., binary) treatments, the groups are defined by treatment arms and the differences define heterogeneous (in x_i) effects. In the continuous case, the grouping is the dimension of heterogeneity and x_i is the treatment. Our results extend naturally to this setting. For a grouping indicator $t_i = 0, \dots, L$, we replace the scalar index in the model (2.1) with $\theta_0(x_i, \mathbf{w}_i, t_i) := \sum_{t=0}^L \mathbb{1}\{t_i = t\} \theta_{0,t}(x_i, \mathbf{w}_i)$, where each $\theta_{0,t}(x_i, \mathbf{w}_i) = \mu_{0,t}(x_i) + \mathbf{w}_i' \boldsymbol{\gamma}_{0,t}$. The level and marginal effect can then be defined groupwise, as $\vartheta_{0,t}(x, \mathbf{w}) = \eta(\theta_{0,t}(x, \mathbf{w}))$ and $\zeta_{0,t}(x, \mathbf{w}) = \eta^{(1)}(\theta_{0,t}(x, \mathbf{w})) \mu_{0,t}^{(1)}(x)$ for some evaluation point $\mathbf{w}$ of control variables.

For example, in a randomized experiment $\vartheta_{0,1}(x, \mathbf{w}) - \vartheta_{0,0}(x, \mathbf{w})$ is the conditional average treatment effect (CATE) function, and the binscatter naturally captures treatment effect heterogeneity along the x_i dimension holding fixed $\mathbf{w}_i = \mathbf{w}$. The rate of change in this heterogeneity is $\zeta_{0,1}(x, \mathbf{w}) - \zeta_{0,0}(x, \mathbf{w})$. Our methods can be used to formally test the null hypothesis that $\vartheta_{0,1}(x, \mathbf{w}) = \vartheta_{0,0}(x, \mathbf{w})$ for all $x \in \mathcal{X}$, which captures the idea of no (heterogeneous) treatment effect. As a second example, our theory can be used to quantify uncertainty for the largest heterogeneous treatment effect: $\hat{x}^* = \arg \sup_{x \in \mathcal{X}} |\hat{\vartheta}_{0,1}(x, \mathbf{w}) - \hat{\vartheta}_{0,0}(x, \mathbf{w})|$. These and many other problems of interest in applied microeconometrics concern the uniform discrepancy of two or more binscatter function estimators, which can be analyzed using our strong approximation and related theoretical results in the supplemental appendix. We do not provide further details here to conserve space, but our software implements several multi-sample estimation, uncertainty quantification, and hypothesis testing procedures.

Figure 4 shows an example of this type of analysis. We divide states into two groups based on their population density, with low and high density states as those with population densities below or above 100 people per square mile, respectively. Density is defined as the average population per square mile, and the data is available from the Census Bureau. Panel (a) shows $\hat{\vartheta}_t(x)$ for each group, i.e. without controls, while panel (b) adds controls and shows $\hat{\vartheta}_t(x, \hat{\mathbf{w}})$, with $\hat{\mathbf{w}}$ set to the sample mean. Note that bin locations differ between the groups because the bias-variance tradeoff is data-dependent. The point estimates show higher uninsured rates in zip codes in low population density states as compared to high density

states. Without controls, there is generally overlap in the confidence bands except for very low incomes. In contrast, when covariates are added, there is a much clearer delineation between the two groups at all but the lowest of income levels. This is made clear in panels (c) and (d), which plot the point estimate of the difference (the CATE) and the associated confidence band, enforcing common binning between the groups. The null hypothesis that $\vartheta_{0,1}(x, \mathbf{w}) = \vartheta_{0,0}(x, \mathbf{w})$ for all $x \in \mathcal{X}$ is rejected in both cases, with test statistics of 7.719 and 8.936, respectively, and negligible p-values. Multi-sample comparisons share the same sensitivity to the chosen evaluation point as discussed above. These issues are unavoidable; researchers must be mindful when implementing the tests and interpreting the results.

6 Conclusion

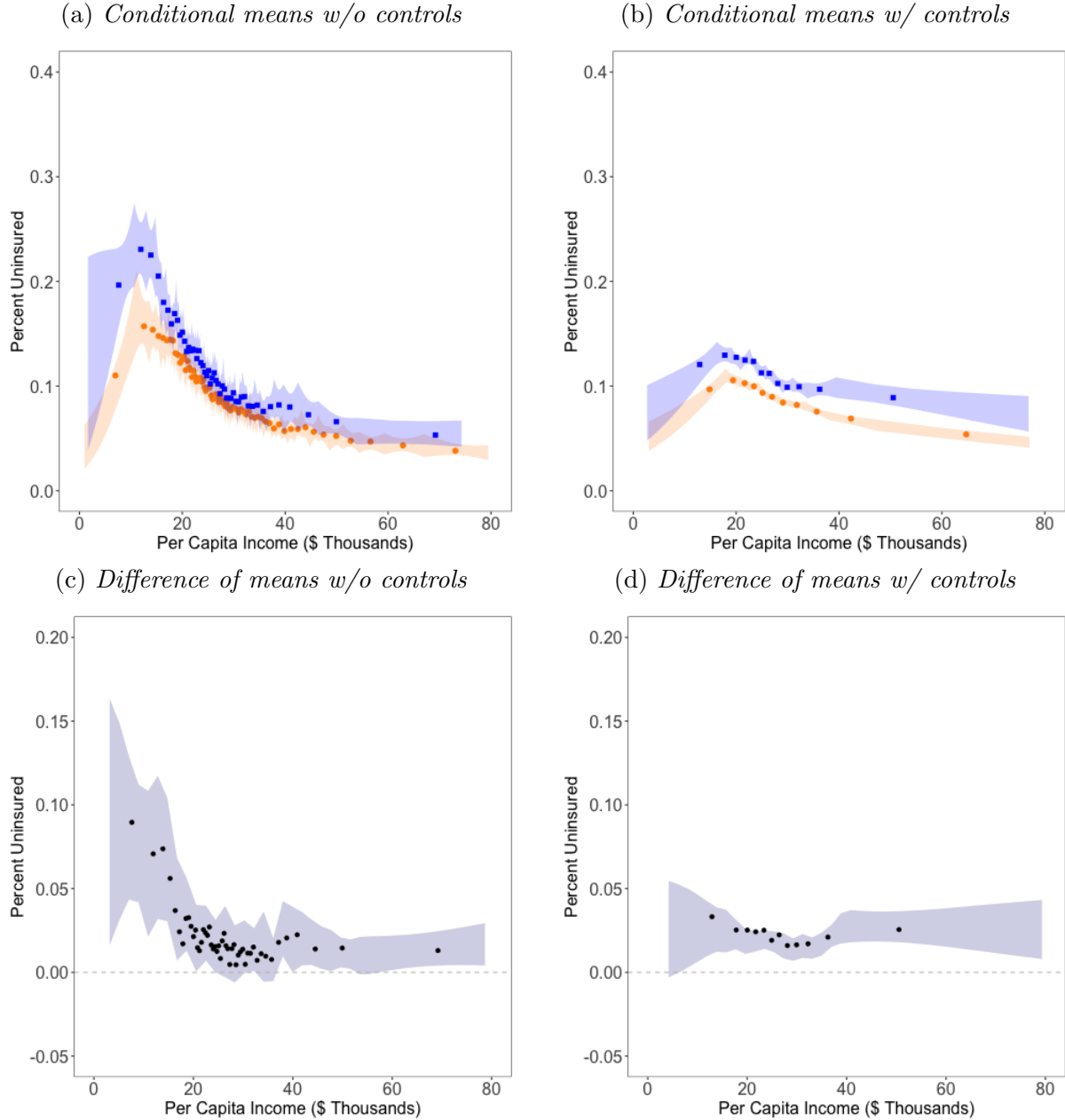
With the rise of large data sets, new visualization tools, such as binned scatter plots, have emerged and gained in popularity. This paper has thoroughly studied binned scatter plots in nonlinear, nonsmooth regression models. Our main contributions are to propose novel nonlinear binscatter methods, together with IMSE-optimal tuning parameter selection and uniform inference methods, including valid confidence bands and functional testing. Our companion `binsreg` software package makes these tools available for applications.

One avenue for future work would be to generalize the analysis beyond a scalar covariate of interest. For example, in two dimensions such an approach would produce “heat maps” which are the bivariate extension of binned scatter plots. Extending our results to that case would be a valuable addition to the practitioner’s toolkit.

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Figure 4: **Two Sample Comparison.** This figure uses the same ACS data to compare areas in low density states (blue) and high density states (orange). Low density states are defined as those with average population per square mile below 100. Panels (a) and (b) show the point estimate (squares or dots) and robust bias corrected confidence band (shaded region) for each group, first without control variables and then with controls added (listed in footnote 4). Panels (c) and (d) show the estimated difference (evaluated using the binning of the low density states) and the associated confidence bands. Shaded regions denote 95% confidence bands and are based on 50,000 random draws.



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A Appendix: Additional Empirical Illustrations

Here we show three additional empirical analyses designed to illustrate the added value of nonlinear and/or nonsmooth modeling in applied contexts. We are grateful to the editor for suggesting these analyses. The discussion and figures below build on several issues raised in the main paper. We stress that these analyses are designed to illustrate methodological issues and not to find novel empirical conclusions.

Robustness. The goal is to estimate a measure of central tendency, similar to Figure 1. Control variables are not used. A linear binscatter (i.e., least squares loss estimation of the conditional mean) may be suitable (footnote 1) but outliers or corrupted data can render it unreliable. This is common in practice and is of particular concern in large data sets. To demonstrate the value of our results in this context, we intentionally corrupt the original data in a controlled way to show how the median binscatter is more robust than least squares, using the unaltered data as a benchmark. Our empirical application is well-suited for this because economic facts and regularities dictate the shape of the conditional mean: the existence of Medicaid implies lower uninsuredness for the poorest areas and cost of insurance implies a decreasing function above certain income levels (see Table 1).

Figure A.1 shows the results of corrupting 32 observations (one tenth of one percent of the data) at low income (Panel (a)) and high income (Panel (b)). This small corruption at low income levels destroys the lower uninsuredness due to Medicaid when fitting the conditional mean using least squares loss, but not when using the conditional median. The same is true at high income levels: the curve is no longer downward sloping when a linear binscatter is applied to the corrupted data. In sum, this analysis shows that the advantages of robust regression methods, long used in parametric modeling, carry over to binscatter contexts.

Predictions outside the support. A different form of robustness is illustrated by nonlinear modeling of the conditional mean with controls. Here we apply (2.3) using both least squares estimation (Example 1) and logistic QMLE (Example 2) with the additional controls of footnote 4. Following the discussion in footnote 1, due to the presence of controls these two

Figure A.1: **Robustness to Corrupted Data or Outliers.** This figure illustrates the robustness of median regression compared to least squares estimation as a measure of centrality. Panels (a) and (b) show, respectively, corruption of data at low and high income levels. In both cases, we corrupt 0.1% of the data into upper outliers.

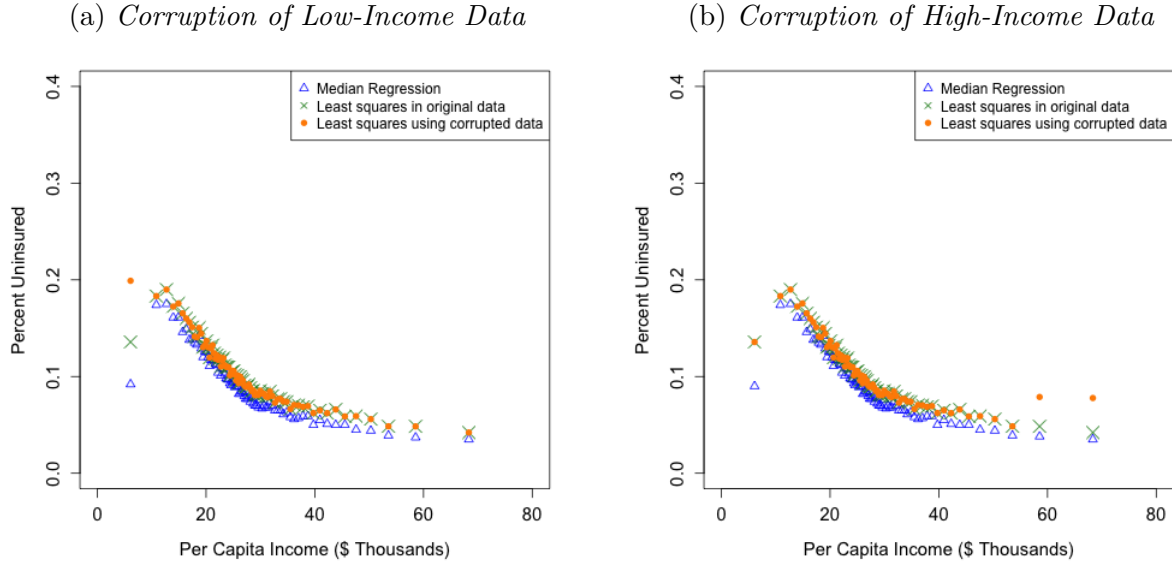
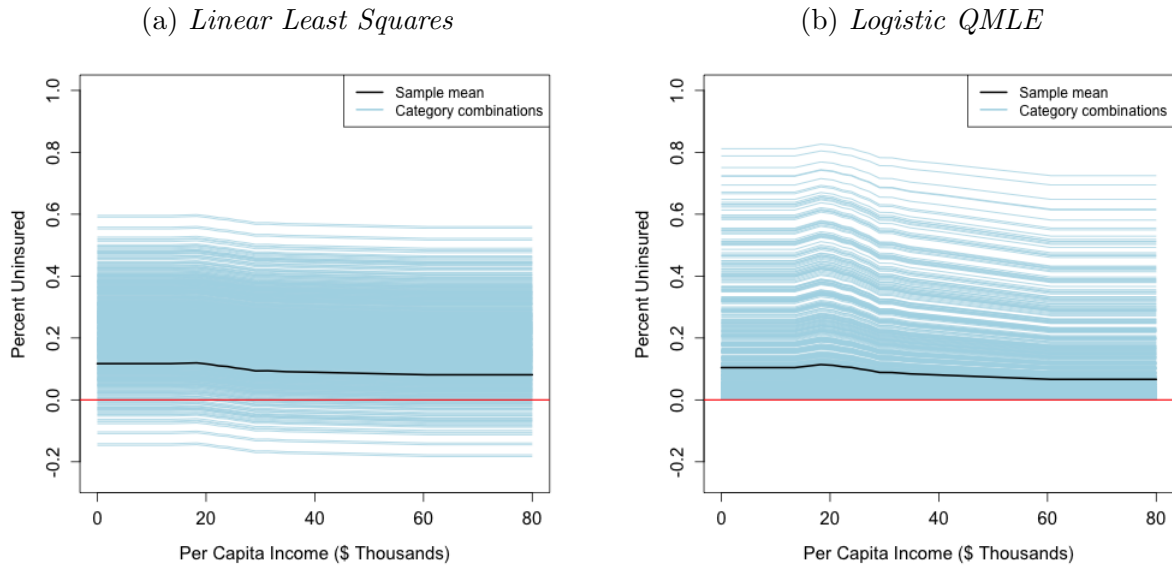


Figure A.2: **Predictions Outside the Unit Interval.** This figure shows that estimating the conditional mean using least squares loss, Panel (a) can lead to fitted values outside the unit interval but logistic QMLE, Panel (b), will not. Changing the evaluation point w results in a shift of the fitted binscatter and least squares does not respect the bounded nature of the outcome. Each light blue line corresponds to a different w , cycling through all possible combinations of realized minima and maxima for the controls. The black line shows the binscatter at the sample mean of w_i .



do not result in identical binscatters as was the case in Figure 1. However, our focus in this analysis is on the evaluation point $\mathbf{w}$ and how this shifts the binscatter. For least squares, this can easily result in predictions outside the unit interval, despite y_i being a fractional outcome. This is similar to the perils of linear probability modeling.

To examine this sensitivity, we create a binscatter plot for each combination of the minimum and maximum values of the coordinates of $\mathbf{w}_i$.⁵ This would correspond, for example, to setting different baseline categories for fixed effects. With nine controls this yields 512 binscatter plots. Figure A.2 shows the results of this exercise, with Panel (a) containing the least squares fits and Panel (b) the logistic QMLE. The light blue curves show each of the 512 fits, in each panel. Here we connect the underlying dots for visual purposes, so that the reader can see each fit individually, though this would not be standard practice. The thicker black line shows, for concreteness, setting $\hat{\mathbf{w}}$ to the sample mean.

The overall shape is not too dissimilar from the two methods, for the sample mean, although the curves are flattened visually due to the expanded y -axis. The important feature is the number of least squares (linear) binscatters which are below zero. Loosely speaking, these curves largely represent small-household areas, high percentages of armed forces, high percentages of residents aged 65 and above, and high percentages of English-only speakers. With aggregate data the identities of these is difficult to interpret, but with individual level data these curves would represent distinct “types” as defined by $\mathbf{w}_i$. Due to the modeling assumptions, the logistic QMLE approach does not suffer from this problem.

Visualizing Uncertainty and Target Parameter. Often the object of interest is defined by a specific loss function. This occurs naturally when visualizing uncertainty. Studying the interquartile range as a measure of dispersion is a typical example. As discussed briefly in the main text, quantile binscatters can be used to evaluate the spread in the conditional distribution more flexibly and richly than reporting single values, just as examining a classic scatter plot can better inform spread than the summary statistics of the 25th and 75th per-

⁵Some of the manufactured $\mathbf{w}$ may lie outside the support $\mathcal{W}$ as required by Assumption 1, but as this is for illustration we abstract from this detail.

centiles alone. This is one motivation for the quantile binscatters (Example 3 and Figure 2). (Quantiles also have policy-relevant reasons for study on their own, as discussed in the text.)

We emphasize that in these cases the target parameter is itself a measure of the spread in the data. The goal is not to visualize the uncertainty of an estimator of, for example, the conditional mean. Although both depend on the variability in the underlying data, they represent different questions. The estimation uncertainty around the mean is relevant for testing hypotheses and making scientific decisions, particularly in causal inference contexts such as identifying most-impacted groups or heterogeneous effects. See Sections 5.2 and 5.3.

Figure A.3 overlays the 10th and 90th percentiles (grey dots) with an estimate of the conditional mean (blue dots) and its 80% confidence band (shaded region). Panel (a) has no controls; Panel (b) adds the nine controls listed in footnote 4. We have little uncertainty regarding the conditional mean, hence the narrow bands. However, there is considerable uncertainty in the outcome y_i given x_i (and $\mathbf{w}_i$ in Panel (b)).

Figure A.3: **Visualizing Uncertainty.** This figure illustrates how quantile regression can be used to visualize uncertainty differently than confidence bands, capturing the spread instead of estimation uncertainty. The grey dots show the 10th and 90th percentiles, the dark blue dots the conditional mean estimate, and the shaded region its associated confidence band.

